NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Sports & Arthroscopy

1. ACL and PCL
2. menisci and cartilage
3. patellofemoral instability
4. rotator cuff
5. shoulder instability
6. elbow in throwers
7. hip arthroscopy and FAI
8. sports medicine and anti-doping

ACL and PCL

1. ACL anatomy and biomechanics

- **Origin:** posteromedial aspect of the **lateral femoral condyle** (medial wall of the lateral condyle, intercondylar notch). Landmarks: **lateral intercondylar ridge ("resident's ridge")** marks the anterior (upper in flexion) limit of the footprint; **lateral bifurcate ridge** separates the AM and PL bundles.
- **Insertion:** tibia, anterior to and between the intercondylar eminences, lateral to the medial tibial spine; **anterior horn of the lateral meniscus** blends with it.
- **Length about 32-38 mm, width about 10-11 mm**. **Two bundles (named by tibial insertion):**
  - **Anteromedial (AM): tight in flexion**; primary restraint to anterior tibial translation (anterior drawer at 90 degrees).
  - **Posterolateral (PL): tight in extension**; resists **rotational** loads (pivot shift) and anterior translation near extension.
- **Blood supply: middle genicular artery** (synovial sheath); **nerve:** posterior articular branches of the **tibial nerve** (mechanoreceptors: proprioception).
- **Function: primary restraint to anterior tibial translation (about 85% at 30 and 90 degrees of flexion)**; secondary restraint to internal rotation and varus/valgus.
- **Strength:** about **2,160 N** ultimate tensile load (native).

2. ACL injury

- **Mechanism: non-contact** (about 70%): **deceleration, cutting, pivoting, landing** with the knee near extension, **valgus and external (or internal) rotation**: "**pop**", immediate swelling (**haemarthrosis: ACL tear in about 70% of acute traumatic haemarthroses** in adults).
- **Females 2-8 times higher risk: narrower notch**, **smaller ACL**, **increased Q angle/valgus**, **quadriceps dominance (less hamstring activation)**, **landing in a more upright position with knee valgus**, hormonal (preovulatory phase), **increased posterior tibial slope** (lateral). **Neuromuscular training programmes** (e.g. **FIFA 11+, PEP**) reduce ACL injury risk.

- **Examination:**
  - **Lachman test** (**most sensitive**; knee at **20-30 degrees** of flexion),
  - **Anterior drawer** (90 degrees; less sensitive acutely, hamstring guarding, posterior horn of medial meniscus acts as a "doorstop"),
  - **Pivot shift** (**most specific**; best under anaesthesia; anterior subluxation of the lateral tibial plateau in extension reducing at **20-40 degrees of flexion** via the **iliotibial band** as it moves posterior to the axis of rotation; **requires an intact IT band and MCL**),
  - **KT-1000 arthrometer** (**side-to-side difference over 3 mm** abnormal).
- **Imaging:**
  - **Radiographs: Segond fracture** (**lateral capsular avulsion of the lateral tibial plateau: anterolateral ligament (ALL)/capsule**: **pathognomonic of ACL tear**), **lateral femoral notch sign** (deep sulcus over 1.5-2 mm in the lateral femoral condyle), tibial spine avulsion (children).
  - **MRI:** discontinuity, **bone bruises ("kissing contusions") on the lateral femoral condyle (sulcus terminalis) and posterolateral tibial plateau** (pivot-shift injury), anterior tibial translation over 5-7 mm, uncovered posterior horn of the lateral meniscus, buckled PCL.
- **Associated injuries:**
  - **Meniscal tears: acute: lateral meniscus more commonly torn** (posterior horn, **root tears**); **chronic ACL deficiency: medial meniscus (posterior horn) tears** increase (it becomes the secondary restraint).
  - **Ramp lesions** (**meniscocapsular separation of the posterior horn of the medial meniscus**: probe through the **posteromedial portal/trans-notch view**; often missed on MRI).
  - **MCL injuries** (commonest associated ligament), **posterolateral corner (PLC)** (**missed PLC = commonest cause of ACL graft failure from a missed injury**), chondral injuries, **"unhappy triad" (O'Donoghue: ACL + MCL + medial meniscus; actually lateral meniscus more often in acute injuries)**.

3. Management of ACL rupture

- **Non-operative:** "copers" (low-demand, no instability, willing to modify activities): **progressive rehabilitation**. **KANON trial (Frobell):**

rehabilitation + optional delayed reconstruction gave similar 2- and 5-year outcomes to early reconstruction in young active adults (**about 50% avoided surgery**). **Increased risk of secondary meniscal tears** with recurrent instability.

- **Operative indications: recurrent instability**, **high-demand pivoting sports**, **repairable meniscal tear**, **combined ligament injuries**, **young age (children: growth-respecting techniques)**, occupational demands.
- **Timing: after resolution of swelling and recovery of full extension and quadriceps control** (avoids **arthrofibrosis**); the "**prehabilitation**" principle. **Early surgery** (within 2-3 weeks) is acceptable when ROM is good; **lock knee from a bucket-handle tear: urgent arthroscopy**.

Graft choice

Graft	Advantages	Disadvantages
Bone-patellar tendon-bone (BPTB)	**Bone-to-bone healing (6-8 weeks), lowest re-rupture in young athletes, strong (about 2,900 N)**	**Anterior knee pain, kneeling pain**, patellar fracture, patellar tendon rupture, **patella baja**, extension loss
Hamstring (semitendinosus ± gracilis, quadrupled)	**Strongest (about 4,000 N quadrupled)**, small incision, **less anterior knee pain**	**Soft-tissue healing (8-12 weeks)**, **tunnel widening**, hamstring weakness (deep flexion), saphenous nerve (infrapatellar branch) injury, **higher re-rupture in young/female athletes with small (under 8 mm) grafts**
Quadriceps tendon (± bone block)	Thick, large cross-section, **low donor morbidity**, increasing use in primary and revision	Quadriceps weakness, patellar fracture (with bone block)
Allograft	No donor morbidity, shorter surgery	**Higher failure in young (under 25), active patients** (especially **irradiated (over 2.5 Mrad)**), slower incorporation, disease transmission (rare), cost

- **Graft diameter: under 8 mm** hamstring grafts have higher revision rates.
- **Graft incorporation: avascular necrosis**, **revascularisation**, **cellular repopulation** and **remodelling (ligamentisation)**: **weakest at about 6-12 weeks**; full maturation 1-2 years.

Technical points

- **Anatomic tunnel placement**: **femoral tunnel low on the lateral wall of the notch (within the native footprint, below the lateral intercondylar ridge)**, drilled via **anteromedial portal**, **outside-in**, or flexible reamers (transtibial drilling often creates a **vertical, non-anatomic graft**: rotational instability with a persistent pivot shift).
- **Commonest technical error: femoral tunnel placed too anteriorly** (graft **tight in flexion**: loss of flexion or graft failure/stretching).
- **Tibial tunnel too anterior**: **roof impingement in extension** (loss of extension, **cyclops lesion** risk); too posterior: vertical graft, PCL impingement.
- **Fixation:** interference screws, suspensory buttons (femoral), staples/screw-post (tibial).
- **Lateral extra-articular tenodesis (LET; modified Lemaire)** or **ALL reconstruction** added in **high-risk patients** (young, high-grade pivot shift, return to pivoting sport, generalised laxity, revision): **STABILITY trial** (hamstring ACLR + LET in young patients: **reduced graft rupture/persistent rotatory laxity**).
- **Double-bundle reconstruction:** no clear clinical advantage over anatomic single-bundle.
- **ACL repair** (proximal avulsion types) is being revisited with augmentation; not standard.

Rehabilitation and return to sport

- **Early full extension, quadriceps activation, closed kinetic chain exercises** (less ACL graft strain than open chain in the early phase); **avoid open-chain quadriceps (knee extension) exercises between 0-45 degrees** early.
- **Return to sport: criteria-based** (strength over 90% limb symmetry index, hop tests, psychological readiness) and **not before 9 months**: each month

delay in return to sport until 9 months reduces re-injury by about 51% (Grindem, Delaware-Oslo).

- **Contralateral ACL injury** risk is similar to graft rupture risk in young athletes.

Complications

- **Arthrofibrosis/loss of extension** (commonest complication; **cyclops lesion** = fibroproliferative nodule anterior to the graft: **loss of terminal extension with a "clunk"**: arthroscopic excision).
- **Graft failure: early (technical: tunnel malposition, fixation failure)**, **late (re-injury)**; **missed PLC/medial injuries**; **increased posterior tibial slope (over 12 degrees)**.
- **Infection** (**coagulase-negative staphylococci**, *S. aureus*; **pre-soak graft in vancomycin** reduces infection), **tunnel widening**, DVT, saphenous nerve injury, anterior knee pain, **osteoarthritis** (higher in meniscectomy cases; reconstruction does not prevent OA).
- **Revision ACLR:** assess **tunnel position, widening (over 15 mm: staged bone grafting)**, **posterior tibial slope (over 12 degrees: slope-reducing osteotomy)**, **varus alignment (HTO)**, missed associated laxities.

4. Paediatric ACL injuries

- **Tibial spine (eminence) fractures** (**Meyers and McKeever classification**: I undisplaced, II hinged anterior elevation, III completely displaced (IIIA, IIIB rotated), IV comminuted): **types I and reducible II = closed reduction in extension, cast**; **III/IV and irreducible II = arthroscopic/open fixation (sutures or screws)**; **entrapment of the intermeniscal ligament or anterior horn of medial meniscus** blocks reduction.
- **Intrasubstance ACL tears in skeletally immature:** early reconstruction (to avoid meniscal/chondral damage) using **physeal-sparing (all-epiphyseal, Micheli-Kocher extraphyseal ITB)** or **transphyseal** techniques (with soft-tissue grafts, small vertical central tunnels, no hardware across the physis) depending on **remaining growth (Tanner stage, bone age)**; risk of **growth disturbance (angular deformity, leg-length discrepancy)**.

5. PCL anatomy and biomechanics

- **Origin: lateral surface of the medial femoral condyle** (anterior notch); **insertion: posterior tibia in a central depression (PCL facet), about 1-1.5 cm below the joint line**.
- **Twice as strong and wider than the ACL** (about 13 mm wide); **two bundles (named by femoral origin): anterolateral (AL, larger, tight in flexion)** and **posteromedial (PM, tight in extension)**.
- **Meniscofemoral ligaments** (from the posterior horn of the lateral meniscus to the medial femoral condyle): **Humphrey (anterior to the PCL)** and **Wrisberg (posterior to the PCL)** (at least one in about 90% of knees).
- **Primary restraint to posterior tibial translation (about 90-95%)**; secondary restraint to external rotation (with PLC).

6. PCL injury

- **Mechanism: dashboard injury** (posteriorly directed force on the proximal tibia with the knee flexed), **fall onto a flexed knee with the foot plantarflexed**, **hyperextension** (combined injuries).
- **Examination:**
  - **Posterior drawer test** (**most sensitive**; 90 degrees of flexion): **grading by tibial step-off** relative to the femoral condyles (**normal about 1 cm anterior step-off**): **grade I: 0-5 mm translation (tibia still anterior to condyles)**, **grade II: 5-10 mm (tibia flush with the condyles)**, **grade III: over 10 mm (tibia posterior to the condyles: suggests combined PCL + PLC/posteromedial injury)**.
  - **Posterior sag (Godfrey) sign**, **quadriceps active test** (contraction reduces posterior subluxation), **reverse pivot shift** (PLC).
  - **Dial test** (external rotation asymmetry over 10 degrees): **at 30 degrees only = isolated PLC**, **at 30 and 90 degrees = PLC + PCL**.
- **Imaging: stress radiographs** (posterior displacement **over 12 mm** side-to-side suggests combined injury), MRI, **avulsion fractures of the tibial PCL insertion**.

Management

- **Isolated grade I-II PCL injuries: non-operative**: **quadriceps-strengthening rehabilitation**, **PCL brace (dynamic anterior drawer force, e.g. Jack brace)**: good functional outcomes; natural history may include **medial compartment and patellofemoral OA** in the long term.
- **Bony avulsions of the PCL tibial insertion: early fixation** (posterior approach (Burks and Schaffer) or arthroscopic) gives excellent results.
- **Grade III, combined injuries (PLC/multiligament), symptomatic chronic instability: reconstruction**: **transtibial** (**"killer turn"** of the graft at the posterior tibia: graft abrasion/attenuation) vs **tibial inlay** (open posterior bone block; avoids the killer turn); **single- (AL bundle)** or **double-bundle**.
- **Risks: popliteal artery injury** during tibial tunnel drilling (**guide-wire penetration; keep the knee flexed at 90 degrees**, use a guard, posteromedial portal to visualise).

7. Posterolateral corner (PLC)

- **Components: lateral (fibular) collateral ligament (LCL)**, **popliteus tendon**, **popliteofibular ligament** (the "big three"), plus lateral capsule, arcuate ligament, biceps femoris, IT band, lateral gastrocnemius.
- **Function:** primary restraint to **varus** and **external rotation** (especially at 30 degrees).
- **Injury:** posterolateral force to the anteromedial tibia, hyperextension/varus; **common peroneal nerve injury in about 15-30%**.
- **Signs: varus thrust gait**, **dial test**, **external rotation recurvatum test**, **posterolateral drawer**, **reverse pivot shift**, **varus stress at 30 degrees (and at 0 degrees: suggests combined cruciate injury)**.
- **Imaging: arcuate sign** (**avulsion of the fibular styloid**: attachments of the popliteofibular, arcuate and fabellofibular ligaments): PLC injury; varus stress radiographs.
- **Treatment: acute repair within about 2-3 weeks** (higher failure of repair alone) or **reconstruction (anatomic, e.g. LaPrade)**: **reconstruction more reliable than repair**; address **varus alignment (HTO first)** in chronic cases; **PLC must be addressed with any cruciate reconstruction** (unrecognised PLC injury causes cruciate graft failure).

8. Multiligament knee injury/knee dislocation

- **Schenck classification: KD I** single cruciate + collateral; **KD II** both cruciates; **KD III M/L** both cruciates + **MCL (M) or PLC (L)**; **KD IV** both cruciates + both collaterals; **KD V** fracture-dislocation; C = arterial injury, N = nerve injury.
- **Vascular injury (popliteal artery)** in about **10-30%** (anterior dislocations: intimal stretch; posterior: transection). **ABI under 0.9: CT angiography/vascular surgery**; **normal pulses do not exclude injury**: serial **ankle-brachial index (ABI)** ± angiography. Ischaemia over 6-8 hours: high amputation rate; **prophylactic fasciotomy** after revascularisation.
- **Peroneal nerve injury** in about 25% (especially **KD III L**).
- **Management:** reduce urgently, assess vascular status, **external fixation** for unstable/vascular/open injuries; **early (within 3 weeks) single-stage or staged ligament repair/reconstruction** generally gives better results than non-operative treatment; **arthrofibrosis** is common.

Exam pearl: **Lachman most sensitive; pivot shift most specific (needs intact ITB/MCL). Segond fracture = ACL tear. Acute ACL: lateral meniscus tears; chronic: medial meniscus. Commonest technical error: femoral tunnel too anterior (tight in flexion). Tibial tunnel too anterior: roof impingement, cyclops lesion.**

Exam pearl: **AM bundle tight in flexion, PL bundle tight in extension (ACL named by tibial insertion); PCL AL bundle tight in flexion (named by femoral origin). Humphrey anterior, Wrisberg posterior to the PCL.**

Exam pearl: **PCL grade III (tibia behind condyles) = suspect PLC. Dial test: 30 only = PLC; 30 and 90 = PLC + PCL. Arcuate sign = fibular styloid avulsion. Missed PLC causes cruciate graft failure. Knee dislocation: check ABI; under 0.9 = angiography.**

Menisci and cartilage

1. Meniscal anatomy

- **Fibrocartilage**: about **70% water**; dry weight mostly **type I collagen (about 90%)**; proteoglycans, cells (**fibrochondrocytes**).
- **Collagen orientation: circumferential fibres (deep, dominant)** convert axial load into **hoop stresses**; **radial ("tie") fibres** resist longitudinal splitting; superficial random mesh.
- **Medial meniscus: C-shaped**, wider posteriorly, **firmly attached to the capsule and deep MCL**: **less mobile (about 2-5 mm excursion)**: **torn more often** (overall). **Lateral meniscus: more circular (O-shaped)**, covers more of the plateau, **more mobile (about 9-11 mm)**; attached to the **popliteus** (popliteal hiatus: no capsular attachment there) and **meniscofemoral ligaments (Humphrey, Wrisberg)**.
- **Blood supply: medial, lateral and middle genicular arteries** via a **perimeniscal capillary plexus**; only the **peripheral 10-30% (medial) and 10-25% (lateral)** is vascular in adults.
  - **Red-red zone** (peripheral, **under 3 mm from the rim**): good healing.
  - **Red-white zone** (**3-5 mm**): moderate healing.
  - **White-white zone** (**over 5 mm**, inner): avascular, poor healing.
- **Function: load transmission** (**about 50% of load in extension and about 85-90% in flexion through the menisci**; **lateral meniscus bears about 70% of lateral compartment load** vs medial about 50%), **shock absorption**, **joint stability** (**posterior horn of the medial meniscus is a secondary restraint to anterior tibial translation**, especially in ACL-deficient knees), **lubrication and nutrition**, proprioception.
- **Total meniscectomy increases contact stresses by about 2-3 times (up to 235%)**: **Fairbank changes** on radiographs after meniscectomy: **ridge (osteophyte) formation, flattening of the femoral condyle, joint space narrowing**. **Lateral meniscectomy** has worse outcomes than medial (lateral compartment convex-on-convex geometry).

2. Meniscal tears

- **Patterns:**

  - **Longitudinal (vertical): bucket-handle** (displaced longitudinal: **locked knee**; **double PCL sign** on MRI, **"fragment in the notch" sign**; commoner medially),
  - **Radial** (transects circumferential fibres: **loss of hoop stress function**; **"ghost meniscus"** sign),
  - **Horizontal (cleavage)** (degenerative; **parameniscal cysts**, especially lateral),
  - **Oblique (flap/parrot-beak)**, **complex/degenerative**,
  - **Root tears** (within about 1 cm of the root attachment): **functionally equivalent to total meniscectomy**; **medial posterior root tears in middle-aged women** with a "pop" on squatting (**meniscal extrusion over 3 mm** on MRI); **lateral posterior root tears with ACL injuries**,
  - **Ramp lesions** (posterior medial meniscocapsular junction; ACL-associated),
  - **Meniscal extrusion** (over 3 mm: root tear or degeneration).
- **Clinical:** joint-line tenderness (**most sensitive sign**), **McMurray test**, **Thessaly test (at 20 degrees)**, **Apley grind**; locking, effusion (delayed, hours).
- **MRI: sensitivity about 90-95%** (medial better than lateral); **grade 3 signal (intrasubstance signal reaching the articular surface)** = tear; grades 1-2 are intrasubstance degeneration (not tears).

3. Management of meniscal tears

- **Degenerative tears in middle-aged/older knees** (with or without OA): **physiotherapy first**; **arthroscopic partial meniscectomy (APM) is no better than sham surgery or physiotherapy** for degenerative tears (**FIDELITY**, **ESCAPE**, Katz **MeTeOR**); APM is reserved for **true mechanical symptoms (locking)** or failure of conservative care.
- **Repair indications ("save the meniscus"): peripheral (red-red/red-white), longitudinal, vertical tears 1-4 cm**, **young patient**, **acute**, **stable knee or with concurrent ACL reconstruction** (**healing rates higher with ACL reconstruction: bleeding, growth factors**), **root tears**, **ramp lesions**, **bucket-handle tears** in young patients (reduce and repair).
- **Repair techniques:**

  - **Inside-out** (**gold standard** for mid-body/posterior tears; vertical mattress sutures; **posteromedial approach (protect the saphenous nerve)**; **posterolateral approach: protect the peroneal nerve (retract biceps femoris posteriorly, work anterior to the lateral gastrocnemius)**),
  - **Outside-in** (anterior horn tears),
  - **All-inside** (posterior horn; device-based; **risk to popliteal neurovascular structures in the lateral posterior horn**),
  - **Vertical mattress sutures are biomechanically strongest** (capture circumferential fibres).
  - **Biological augmentation:** marrow venting (notch microfracture), **fibrin clot**, rasping/trephination.
- **Root repair: transtibial pull-out** repair or suture anchor; slows extrusion and OA progression vs partial meniscectomy (in knees without advanced OA).
- **Rehabilitation after repair:** protected weight bearing/limited deep flexion (**avoid squatting beyond 90 degrees for about 3-4 months**).
- **Outcomes: repair success about 70-90%** (better laterally, with ACL reconstruction, acute, peripheral); **failure rate higher medially and in isolated repairs**.
- **APM outcomes:** good for symptomatic unstable flaps in younger knees, but **long-term OA risk** proportional to the amount removed.

4. Discoid meniscus

- **Lateral (almost always)**; **about 3-5% of the population (more in Asians, up to about 15%)**; **bilateral in about 20%**.
- **Watanabe classification:**
  - **Type I: complete** (covers the entire plateau),
  - **Type II: incomplete** (covers under 80%),
  - **Type III: Wrisberg variant** (**no posterior meniscotibial (coronary) attachment**, only the **meniscofemoral ligament of Wrisberg**: **hypermobile**: **"snapping knee syndrome"** in children).
- **Imaging:** radiographs: **widened lateral joint space**, squared lateral condyle, cupping of the lateral plateau; **MRI: three or more contiguous 5-**

mm sagittal slices showing a continuous "bow-tie" (meniscal body continuity), or **transverse diameter over 15 mm or over 20% of tibial width**.

- **Treatment: asymptomatic: observe**; **symptomatic: arthroscopic saucerisation (partial meniscectomy to a rim of about 6-8 mm)** ± **repair of peripheral instability (Wrisberg type)**; **avoid total meniscectomy** (early lateral OA, osteochondritis dissecans of the lateral femoral condyle).

5. Meniscal allograft transplantation (MAT) and scaffolds

- **Indication: young (under about 50), symptomatic post-meniscectomy knee** (pain in the meniscus-deficient compartment) **with minimal arthritis (Outerbridge under III)**, **normal alignment (or corrected by osteotomy)**, **stable knee (or concomitant ACL reconstruction)**, **BMI under about 35**.
- **Contraindications:** advanced OA, uncorrected malalignment/instability, inflammatory arthritis, obesity.
- **Size-matched fresh-frozen allograft** (radiographic sizing: **Pollard method**); fixation with **bone plugs (medial) or a bone bridge/keyhole (lateral: horns close together)**, or soft-tissue sutures.
- **Outcomes: about 70-80% survival at 10 years**; better laterally; **shrinkage, extrusion, tears** common.
- **Collagen (CMI) and polyurethane (Actifit) scaffolds** for partial segmental defects (intact rim needed).

6. Articular cartilage: structure

- **Hyaline cartilage: about 65-80% water**; **type II collagen (about 90-95% of collagen; tensile strength)**; **proteoglycans (aggrecan: GAG chains of chondroitin and keratan sulphate bound to hyaluronic acid by link protein)**: **negative charge attracts water: compressive stiffness (swelling pressure)**; **chondrocytes (about 1-5% of volume)**: produce matrix; **avascular, aneural, alymphatic**: **nutrition by diffusion from synovial fluid**: poor healing.
- **Other collagens: type IX** (cross-links type II), **type X (calcified zone, hypertrophic chondrocytes)**, **type XI** (regulates fibril size), **type VI** (pericellular).
- **Zones:**

  - **Superficial (tangential):** collagen **parallel** to the surface (**highest collagen, resists shear**), **lowest proteoglycan**, flattened cells, **lamina splendens**, **highest water content**,
  - **Middle (transitional):** oblique collagen, round cells,
  - **Deep (radial):** collagen **perpendicular** to the surface, **highest proteoglycan, lowest water**, columns of cells (resists compression),
  - **Tidemark** (separates uncalcified from calcified cartilage),
  - **Calcified cartilage** (type X collagen; anchors to subchondral bone).
- **Lubrication: boundary lubrication (lubricin: proteoglycan 4)**, **fluid-film**, **weeping/elastohydrodynamic**; **biphasic** (solid + fluid) viscoelastic material.
- **Ageing vs OA:**

Parameter	Ageing	Osteoarthritis
Water content	**Decreases**	**Increases (early)**
Collagen	Unchanged/cross-linking increases	Network disrupted
Proteoglycan content	Decreases	**Decreases**
Chondroitin sulphate	**Decreases**	**Increases (chondroitin 4-sulphate)**
Keratan sulphate	**Increases**	**Decreases**
Modulus of elasticity	**Increases (stiffer)**	**Decreases**
Degradative enzymes (**MMP-1, -13, ADAMTS-4/5**, IL-1, TNF)	Normal	**Increased**
Chondrocyte number	Decreases; size increases	Early proliferation (clusters/cloning)

7. Cartilage injury and repair

- **Partial-thickness injuries (not crossing the tidemark): do not heal** (no access to marrow cells).
- **Full-thickness (through the subchondral plate):** marrow cells form **fibrocartilage (type I collagen)**: mechanically inferior to hyaline.

- **Classification: Outerbridge** (grade I softening; **II fissuring/fragmentation under 1.5 cm (0.5 inch) diameter**; **III over 1.5 cm**; **IV exposed subchondral bone**), **ICRS** (0 normal; 1 superficial; 2 **under 50% depth**; 3 **over 50% depth (to calcified layer)**; 4 to subchondral bone).
- **Treatment algorithm for focal chondral defects** (young patients, **correct alignment, stability and meniscal deficiency first**):

Defect	Low-demand	High-demand
Small (under 2 cm^2)	**Debridement/chondroplasty**, **microfracture/marrow stimulation**	**Osteochondral autograft transfer (OATS/mosaicplasty)**, microfracture
Large (over 2-4 cm^2)	Microfracture (less durable), **ACI/MACI**	**Autologous chondrocyte implantation (ACI/MACI)**, **osteochondral allograft (OCA)** (especially with bone loss)

- **Microfracture (Steadman): debride to a stable vertical rim**, **remove the calcified cartilage layer**, awls **3-4 mm apart, 2-4 mm deep**; **fibrocartilage**; **protected weight bearing and CPM for 6-8 weeks**; **results deteriorate after about 2-5 years**, best in **young (under 40), small (under 2-4 cm^2), femoral condyle** lesions; **intralesional osteophytes (subchondral bone overgrowth)** complicate later ACI.
- **OATS (osteochondral autograft): hyaline cartilage** transfer from **less weight-bearing areas** (**lateral trochlea, intercondylar notch**); **defects under about 2 cm^2**; **donor site morbidity** (patellofemoral pain); plugs must be **flush** (**proud plugs: high contact pressures**; recessed: fibrocartilage overgrowth).
- **ACI/MACI: two-stage** (**arthroscopic biopsy of cartilage**, culture, **implantation at about 4-6 weeks** under a **periosteal patch (first generation: hypertrophy commonest complication)**, **collagen membrane (second)** or **matrix-seeded (MACI, third generation)**); **hyaline-like** repair tissue; **larger defects (2-10 cm^2)**; better than microfracture for larger defects in some trials; **slow rehabilitation (about 12 months)**.
- **Osteochondral allograft (fresh, refrigerated, used within about 28 days)**: **large defects with subchondral bone loss**, **OCD**, AVN, salvage;

viable chondrocytes (over 70% viability required); **no immune rejection of cartilage (chondrocytes protected in matrix)**, bone portion incorporates via **creeping substitution**.

- **Subchondral insufficiency fracture of the knee (SIFK, formerly SONK): elderly women**, **medial femoral condyle**, often after a **medial meniscus posterior root tear**; **sudden pain**; MRI: **subchondral low-signal line with bone marrow oedema**; protected weight bearing; UKA/TKA if collapse.

8. Osteochondritis dissecans (OCD)

- **Subchondral bone lesion** (repetitive microtrauma, vascular, genetic) with **risk of instability of the overlying cartilage**; **lateral aspect of the medial femoral condyle (about 70%)** (**Wilson sign**: pain at 30 degrees of flexion with internal rotation, relieved by external rotation).
- **Juvenile OCD (open physes)** has a **much better prognosis** (**about 50-75% heal with non-operative care**: activity restriction 3-6 months) than **adult OCD** (closed physes: usually need surgery).
- **MRI signs of instability (De Smet/Kijowski): high T2 signal line behind the fragment**, **cysts** beneath the lesion, **articular fracture line**, **fluid-filled osteochondral defect**.
- **Treatment:**
  - **Stable lesions in juveniles: activity modification, immobilisation**; **retrograde/antegrade drilling** if not healing after 3-6 months.
  - **Unstable but salvageable: fixation** (**bioabsorbable or metal compression screws, chondral darts**, bone pegs).
  - **Unsalvageable/loose bodies: excision + cartilage restoration (OATS, OCA, ACI)**.
- **Other sites: capitellum** (**throwers/gymnasts**: see elbow notes), **talar dome** (**posteromedial (deeper, cup-shaped, atraumatic)** and **anterolateral (shallow, wafer-shaped, traumatic)**).

Exam pearl: **Hoop stresses via circumferential fibres; radial and root tears abolish them (root tear = total meniscectomy). Red-red zone (under 3 mm) heals; repair with ACL reconstruction heals best; vertical mattress strongest. Degenerative tears: physio first (FIDELITY, ESCAPE).**

Exam pearl: **Discoid lateral meniscus: Watanabe III (Wrisberg) is hypermobile: saucerise + repair. MAT: young, meniscectomised, minimal OA, aligned, stable.**

Exam pearl: **Cartilage: type II collagen, aggrecan; superficial zone parallel collagen/highest water; deep zone highest proteoglycan. Ageing: water down, stiffer, keratan up; OA: water UP, keratan down, chondroitin up, MMP-13/ADAMTS up. Microfracture: fibrocartilage (type I); OATS: hyaline, small; ACI/MACI: larger defects; OCA: large osteochondral defects.**

Exam pearl: **OCD: lateral aspect of the medial femoral condyle; juvenile heals non-operatively more often; high T2 line behind the fragment = unstable. SIFK: elderly women, medial femoral condyle, medial root tear.**

Patellofemoral instability

1. Anatomy and biomechanics

- **Patella:** the **largest sesamoid**; **thickest articular cartilage in the body (up to about 5-7 mm)**; **lateral facet larger** than the medial; **odd facet** (medial, contacts in deep flexion).
- **Function: increases the moment arm of the quadriceps (by about 30-50%)**, centralises forces, protects the knee.
- **Contact areas:** at **20 degrees**, the **distal pole** contacts the trochlea; contact moves **proximally** on the patella with increasing flexion; **maximum contact area at about 90 degrees**; **at about 120-135 degrees**, the odd facet contacts the medial femoral condyle.
- **Patellofemoral joint reaction force: about 0.5 x body weight walking**, **3-4 x stairs**, **7-8 x squatting**; increases with flexion (and with open-chain extension near full extension: high quadriceps force).
- **Engagement:** the patella is **laterally positioned in extension** and **engages the trochlear groove at about 20-30 degrees of flexion**: **instability occurs in early flexion (0-30 degrees)**, where the **soft-tissue restraints matter most**.

Restraints to lateral patellar translation

- **Medial patellofemoral ligament (MPFL): primary restraint (about 50-60%) at 0-30 degrees**. **Femoral attachment (Schöttle point): between the adductor tubercle and the medial epicondyle** (just anterior to the posterior cortex extension line, proximal to the posterior origin of the medial condyle on a true lateral radiograph); **patellar attachment: proximal two-thirds of the medial border**. **Injured in almost all (about 90-100%) first-time dislocations**, commonly at the **femoral attachment** (adults) or **patellar attachment** (children).
- **Medial patellotibial ligament, medial patellomeniscal ligament, medial retinaculum** (secondary, more important in deeper flexion), **vastus medialis obliquus (VMO)** (dynamic).
- **Trochlear geometry**: the main restraint **beyond about 30 degrees** of flexion.

2. Risk factors for instability (Dejour "major instability factors")

1. **Trochlear dysplasia** (**most important; about 85-96% of patients with recurrent instability**).
2. **Patella alta** (**delays engagement of the patella in the trochlea**).
3. **Increased tibial tubercle-trochlear groove (TT-TG) distance** (**over 20 mm abnormal**; normal about 10-13 mm; lateralised tubercle).
4. **Patellar tilt** (over 20 degrees on CT).
5. **Other factors: generalised ligamentous laxity (Beighton score)**, **femoral anteversion** (increased), **external tibial torsion**, **genu valgum (increased Q angle)**, **VMO dysplasia**, **female sex**, **young age (adolescents)**, **previous dislocation**, family history.
6. **Q angle:** line from ASIS to the patellar centre and from the patellar centre to the tibial tubercle: **normal about 14 degrees (men) and about 17 degrees (women)**; measured in extension (and in 30 degrees flexion to seat the patella).

3. Imaging and measurements

Radiographs

- **True lateral (posterior condyles superimposed):**
  - **Crossing sign** (**trochlear floor line crosses the anterior border of the lateral condyle**: **trochlear dysplasia**: the trochlea becomes flat at that point),
  - **Supratrochlear spur (bump)**,
  - **Double contour** (hypoplastic medial facet).
  - **Patellar height indices:**
  - **Insall-Salvati ratio** (**patellar tendon length / patellar length**; **normal 0.8-1.2**; **over 1.2 = alta**, under 0.8 = baja),
  - **Caton-Deschamps index** (distance from the **inferior pole of the patellar articular surface to the anterosuperior tibial plateau / articular surface length**; **normal 0.6-1.2**; **over 1.2 = alta**; **not affected by tibial tubercle osteotomy**: preferred for planning distalisation),
  - **Blackburne-Peel ratio** (perpendicular distance from the tibial plateau line to the inferior articular pole / articular length; **over 1.0 = alta**).

- **Axial (Merchant view, 45 degrees knee flexion): sulcus angle (normal about 138 ± 6 degrees; over 145 = dysplasia)**, **congruence angle (normal -6 ± 11 degrees; over +16 abnormal: lateral subluxation)**, **lateral patellofemoral angle** (Laurin: should open laterally).

CT/MRI

- **TT-TG distance** (**over 20 mm = abnormal**; MRI values are smaller than CT by about 2-4 mm), **TT-PCL distance (over 24 mm)** (less affected by trochlear dysplasia/knee rotation).
- **Trochlear dysplasia (Dejour classification):**
  - **Type A: crossing sign**; shallow trochlea (sulcus angle over 145 degrees) but preserved morphology,
  - **Type B: crossing sign + supratrochlear spur**; flat or convex trochlea,
  - **Type C: crossing sign + double contour**; asymmetric facets (convex lateral, **hypoplastic medial facet**),
  - **Type D: crossing sign + spur + double contour**; **"cliff" pattern** (vertical step between facets).
- **Patellar tilt**, **femoral anteversion** and **tibial torsion** (rotational profile CT), **MRI**: **MPFL tear**, **bone bruise** (**medial patellar facet and lateral femoral condyle**: pathognomonic of transient dislocation), **osteochondral fractures/loose bodies**.

4. Acute first-time patellar dislocation

- **Mechanism: twisting on a flexed, valgus knee with the foot planted** (internal rotation of the femur on a fixed tibia), or direct blow; usually **lateral** dislocation; often reduces spontaneously with knee extension.
- **Examination:** haemarthrosis, **tenderness over the medial retinaculum/MPFL femoral attachment (Bassett sign)**, **apprehension sign (Fairbank)**, **patellar glide (quadrants: over 2-3 quadrants lateral translation = laxity)**.
- **Osteochondral fractures** in about **25-50%** (**inferomedial patellar facet** (commonest) or **lateral femoral condyle**): radiographs/MRI.
- **Management:**
  - **Non-operative for most first-time dislocations** without loose bodies: **brief immobilisation for comfort**, **early ROM, quadriceps (VMO) and**

hip abductor/external rotator strengthening, **patellar stabilising brace/taping**.

- **Surgery after the first dislocation if: osteochondral fracture/loose body** (**fix or remove: arthroscopy**), **substantial MPFL disruption with lateral patellar subluxation on the axial view (relative)**, **high-risk patients** (young with **dysplasia + patella alta + open physes: recurrence risk over 50-70%**).
- **Recurrence after the first dislocation: about 15-45% overall** (**up to about 70%** in skeletally immature with dysplasia).

5. Recurrent patellar instability: surgical options

- **Principle: address the predominant anatomical abnormalities ("à la carte" surgery, Dejour).**

MPFL reconstruction

- **Workhorse procedure** for recurrent instability: **gracilis or semitendinosus autograft**; femoral fixation at the **anatomic Schöttle point** (**fluoroscopic confirmation**); **graft fixed at about 30-60 degrees of flexion** with the patella centred in the trochlea; **avoid over-tensioning** (**medial patellofemoral overload, pain, loss of flexion, medial instability**).
- **Femoral tunnel malposition: too proximal** (graft **tight in flexion**: **loss of flexion and medial patellar overload**), **too distal** (tight in extension/loose in flexion).
- **Paediatric: femoral attachment is just distal to the distal femoral physis**: drill below the physis or use **physeal-sparing (soft-tissue/adductor sling) techniques**.
- **Complications: patellar fracture** (transverse tunnels across the patella: use **anchors or short sockets**), recurrent instability, stiffness, overconstraint.
- **MPFL repair** has higher recurrence than reconstruction.

Tibial tubercle osteotomy (TTO)

- **Medialisation (Elmslie-Trillat)**: **increased TT-TG (over 20 mm)**: **aim for TT-TG about 10-15 mm**; **avoid over-medialisation** (increases medial tibiofemoral and **medial patellofemoral** contact pressures).

- **Anteromedialisation (Fulkerson)**: **oblique osteotomy**: medialises + **anteriorises (unloads the distal and lateral patellar facets)**: **lateral facet or distal patellar chondral lesions** with instability/malalignment; **contraindicated for proximal/medial facet lesions** (shifts load onto them).
- **Distalisation**: **patella alta (Caton-Deschamps over about 1.2-1.3)**: target about 1.0.
- **Complications: proximal tibial fracture** (**avoid early full weight bearing with Fulkerson: risk of tibial fracture especially at 6-12 weeks**), non-union, **compartment syndrome**, **DVT**, **anterior tibial artery/deep peroneal nerve injury (posterior cortex drilling)**, painful hardware, **Hauser procedure (posteromedialisation: increases PFJ pressure: abandoned; causes PFJ arthritis)**.
- **Roux-Goldthwait** (lateral half of the patellar tendon transferred medially) in skeletally immature (avoids the tibial tubercle apophysis: **TTO contraindicated with open physes: genu recurvatum**).

Trochleoplasty

- **Severe trochlear dysplasia (Dejour B and D with a supratrochlear spur over 5 mm)**, especially **revision/failed MPFL reconstruction**: **sulcus-deepening trochleoplasty (Dejour/Bereiter)**, **recession wedge trochleoplasty** (for spur). **Complications:** arthrofibrosis, cartilage necrosis, PFJ arthritis; **contraindicated with open physes and established PFJ arthritis**.

Lateral retinacular release

- **Never as an isolated procedure for instability** (no benefit, can cause **iatrogenic medial patellar instability**). **Lateral lengthening** for a **tight lateral retinaculum with a positive patellar tilt test (tilt under 0 degrees: cannot lift lateral facet to neutral)**.

Rotational/angular osteotomies

- **Femoral derotational osteotomy** for **excessive femoral anteversion (over about 25-30 degrees)**; **distal femoral varus osteotomy** for **genu valgum** (over about 5 degrees mechanical valgus) in recurrent instability; **tibial derotation** for external tibial torsion.

6. Special situations

- **Habitual (obligatory) dislocation:** dislocates with **every flexion**, in children: **quadriceps contracture** (**vastus lateralis/iliotibial band contracture**; prior quadriceps injections): **quadricepsplasty/lengthening with realignment**.
- **Congenital (permanent) dislocation:** irreducible from birth; associated with **Down syndrome, nail-patella syndrome (LMX1B: hypoplastic patella, iliac horns, renal disease)**, **arthrogryposis**: early soft-tissue realignment.
- **Medial patellar instability: iatrogenic after lateral release** (vastus lateralis tendon detachment); **lateral retinaculum reconstruction/repair**.
- **Patellofemoral pain syndrome (anterior knee pain): diagnosis of exclusion**; **hip abductor/external rotator and quadriceps strengthening** (best evidence), taping, orthoses; **no surgery**.
- **Patellar tendinopathy (jumper's knee): eccentric/heavy slow resistance exercise**, **decline squats**; **Blazina staging**; surgery (debridement) for refractory cases.
- **Sinding-Larsen-Johansson** (inferior pole traction apophysitis) and **Osgood-Schlatter** (tibial tubercle traction apophysitis): self-limiting; **excise the ossicle** after maturity if symptomatic.

7. Summary: planning algorithm

Pathology	Threshold	Procedure
Torn/insufficient MPFL	Recurrent instability	**MPFL reconstruction (all cases)**
TT-TG increased	**Over 20 mm**	**Medialising TTO** (± anteriorisation for lateral/distal facet wear)
Patella alta	**Caton-Deschamps over 1.2**	**Distalising TTO** (or MPFL alone if mild)
Severe trochlear dysplasia	**Dejour B/D, spur over 5 mm**	**Trochleoplasty** (revision mainly)
Femoral anteversion	**Over about 25-30 degrees**	**Femoral derotation osteotomy**

Pathology	Threshold	Procedure
Valgus	Mechanical valgus over about 5 degrees	**Distal femoral varus osteotomy**
Tight lateral retinaculum	Tilt test negative	**Lateral lengthening** (never isolated release)

Exam pearl: **MPFL = primary restraint 0-30 degrees; femoral attachment (Schöttle point) between the adductor tubercle and medial epicondyle; torn in nearly all dislocations. Dejour major factors: trochlear dysplasia (crossing sign), patella alta, TT-TG over 20 mm, tilt over 20 degrees.**

Exam pearl: **Bone bruise on the medial patellar facet and lateral femoral condyle = transient patellar dislocation. Osteochondral fracture of the inferomedial patella: fix or remove. First-time dislocation without loose body: non-operative.**

Exam pearl: **Proximal femoral tunnel in MPFLR = tight in flexion, medial overload. Fulkerson AMZ unloads lateral/distal facets (avoid for proximal/medial lesions). Caton-Deschamps index is used for distalisation. Never isolated lateral release (iatrogenic medial instability). TTO contraindicated with open physes (recurvatum): Roux-Goldthwait.**

Rotator cuff

1. Anatomy

Muscle	Nerve (root)	Insertion	Action	Test
Supraspinatus	**Suprascapular (C5-6)**	Greater tuberosity (superior facet)	**Abduction initiation**, depresses/compresses the head	**Jobe (empty can) test**, drop arm test
Infraspinatus	**Suprascapular (C5-6)**	Greater tuberosity (middle facet)	**External rotation (with arm at side)**	**External rotation lag sign**, resisted ER
Teres minor	**Axillary (C5-6)**	Greater tuberosity (inferior facet)	**External rotation in abduction**	**Hornblower's sign** (ER in 90 degrees abduction), **drop sign**
Subscapularis	**Upper and lower subscapular (C5-6)**	**Lesser tuberosity**	**Internal rotation**, anterior stability	**Lift-off (Gerber)**, **belly-press (Napoleon)**, **bear-hug** (most sensitive for upper subscapularis), **internal rotation lag sign**

- **Rotator cable** (crescent-shaped thickening perpendicular to the cuff fibres, from the biceps to the infraspinatus insertion): **"suspension bridge"** load transfer: **crescent tears** (within the cable) can remain functional.
- **Footprint:** supraspinatus about **12-25 mm** medial-lateral; **"critical zone" (Codman)**: **hypovascular area about 1 cm proximal to the supraspinatus insertion** (controversial).

- **Force couples: coronal plane: deltoid vs inferior cuff (infraspinatus, teres minor, subscapularis)**; **transverse plane: subscapularis vs infraspinatus/teres minor**. Tears that **disrupt a force couple** (e.g. **posterosuperior tears extending into the infraspinatus and teres minor**) cause **pseudoparalysis**.
- **Long head of biceps (LHB):** from the **supraglenoid tubercle and superior labrum**; runs in the **bicipital groove** stabilised by the **biceps pulley (SGHL + CHL)** and **transverse humeral ligament**; **subscapularis tears are associated with medial LHB subluxation**.

2. Pathogenesis

- **Intrinsic (degenerative): age** (prevalence of asymptomatic full-thickness tears rises with age: **about 25% over 60 and about 50% over 80**), **hypovascularity**, **smoking**, **diabetes**, **hypercholesterolaemia**, genetic.
- **Extrinsic: subacromial impingement (Neer)**: **acromial morphology (Bigliani: type I flat, II curved (commonest), III hooked: associated with tears)**, **critical shoulder angle (CSA over 35 degrees: cuff tears; under 30 degrees: OA)**, **os acromiale**, **acromial spurs**; **internal impingement** (**throwers: posterosuperior glenoid contacts the undersurface of the posterior supraspinatus/anterior infraspinatus in abduction-external rotation**: **articular-sided partial tears + posterosuperior labral lesions**, associated with **GIRD**).
- **Traumatic tears** (younger patients, falls, dislocations in over 40s).

3. Classification

- **Partial-thickness tears: Ellman**: **A (articular), B (bursal), C (interstitial)**; **grade 1 under 3 mm, grade 2 3-6 mm, grade 3 over 6 mm (over 50% of tendon thickness, about 12 mm)**. **PASTA lesion** = **partial articular supraspinatus tendon avulsion** (commonest partial pattern).
- **Full-thickness tear size (DeOrio and Cofield): small under 1 cm, medium 1-3 cm, large 3-5 cm, massive over 5 cm**. **Gerber definition of massive tear: complete tear of two or more tendons**.
- **Tear shape (Burkhart/Ellman): crescent** (mobile medial-lateral: direct repair to bone), **U-shaped** (**margin convergence (side-to-side) first**, then repair to bone), **L-shaped**, **massive contracted immobile**.

- **Patte classification of retraction: stage 1** close to the insertion, **stage 2** at the humeral head, **stage 3** at the glenoid.
- **Goutallier classification of fatty infiltration (CT, adapted to MRI by Fuchs): 0 normal; 1 some fatty streaks; 2 more muscle than fat; 3 equal fat and muscle; 4 more fat than muscle**. **Grade 3-4 (over 50% fat): poor outcome, high re-tear, irreversible**.
- **Tangent sign** (supraspinatus does not cross a line drawn between the superior borders of the coracoid and the scapular spine on sagittal oblique MRI): **atrophy**.
- **Hamada classification** (cuff tear arthropathy; see shoulder arthroplasty).
- **Subscapularis tears (Lafosse):** I partial upper third; II complete upper third; III complete upper two-thirds; IV complete with head centred; V complete with eccentric head (subluxation).

4. Clinical evaluation

- **Pain** (night pain, lateral arm/deltoid region), **weakness**, **painful arc (60-120 degrees)**.
- **Impingement tests: Neer sign** (passive forward flexion with internal rotation) and **Neer test** (pain relieved by subacromial lidocaine injection), **Hawkins-Kennedy test** (forward flexion 90 degrees + internal rotation).
- **Strength tests:** see table; **external rotation lag sign** (infraspinatus), **hornblower (teres minor)**, **drop arm (Codman)**.
- **Pseudoparalysis**: **active forward elevation under 90 degrees with full passive range** (massive tears).
- **Imaging:**
  - **Radiographs: acromiohumeral interval (AHI under 7 mm suggests a massive cuff tear; under 5 mm = irreparable)**, **greater tuberosity sclerosis/cysts**, **acromial morphology (outlet view)**, **critical shoulder angle**, cuff tear arthropathy changes.
  - **Ultrasound** (dynamic, cheap; operator-dependent; similar accuracy to MRI for full-thickness tears).
  - **MRI** (tear size, retraction, **atrophy/fatty infiltration**, LHB, labrum); **MR arthrography** (articular-sided partial tears).

5. Management

Subacromial pain (impingement) without full-thickness tear

- **Physiotherapy (scapular and cuff strengthening), NSAIDs, subacromial steroid injection.**
- **Subacromial decompression (acromioplasty): no better than placebo/arthroscopy only (CSAW, FIMPACT trials)**: limited role.

Partial-thickness tears

- **Non-operative first**; **under 50%**: **debridement ± acromioplasty**; **over 50% thickness (Ellman 3)** or bursal-sided over 25-30%: **repair** (**transtendon repair for PASTA** or **completion and repair**).

Full-thickness tears

- **Acute traumatic tears in active patients: early repair (within about 3-6 weeks)** (before retraction and fatty infiltration).
- **Chronic degenerative tears: trial of physiotherapy** (successful in **about 75%** in some studies), **repair if persistent symptoms**, especially **younger, active patients**, with **reparable tears (good muscle quality: Goutallier under 3)**.
- **Natural history of untreated full-thickness tears: about 50% enlarge** (over years; **symptomatic tears, larger tears and those involving the anterior cable are more likely to progress**).

Repair techniques

- **Arthroscopic** (standard) vs **mini-open** vs **open** (similar outcomes).
- **Single-row** vs **double-row** (**double-row: better footprint restoration, higher biomechanical strength, lower re-tear rates for larger tears**, but similar clinical outcomes overall) vs **transosseous-equivalent (suture-bridge)** (**most widely used for medium-large tears**).
- **Margin convergence** for U-shaped tears (reduces strain on the repair).
- **Biceps tenotomy or tenodesis** if the LHB is diseased/unstable (**tenotomy: Popeye deformity, cramping**; **tenodesis: younger/cosmetic concern**).
- **Rehabilitation: sling for about 4-6 weeks** (passive ROM early or delayed: **similar outcomes; early motion may increase re-tear in large tears**),

active at about 6 weeks, strengthening at about **12 weeks**; tendon-bone healing about **12-16 weeks**.

Outcomes and re-tears

- **Re-tear rates: about 10-20% for small/medium tears, up to 40-90% for massive tears** (MRI); **re-tears often still have good pain relief**, though **healed repairs have better strength**.
- **Risk factors for re-tear: age over 65**, **large tear size**, **retraction**, **fatty infiltration (Goutallier 2+)**, **smoking**, **diabetes**, osteoporosis, **delayed repair**, poor tendon quality, **workers' compensation**.
- **Re-tear pattern: Cho type 1 (at the footprint: commonest with single-row)** and **type 2 (at the musculotendinous junction, medial to the medial row: double-row)**.

6. Massive irreparable cuff tears

- **Options (age, pseudoparalysis, arthritis, subscapularis and teres minor status):**
  - **Non-operative**: deltoid rehabilitation (anterior deltoid programme).
  - **Arthroscopic debridement ± biceps tenotomy** (pain relief in low-demand elderly; **preserve the coracoacromial arch**: **avoid acromioplasty/CA ligament release** to prevent **anterosuperior escape**).
  - **Partial repair** (restore force couples: repair infraspinatus/subscapularis).
  - **Superior capsular reconstruction (SCR)** (**Mihata**: **fascia lata autograft (thick, 6-8 mm)** or **dermal allograft** from the superior glenoid to the greater tuberosity: restores **superior stability**; **young patients without arthritis, intact/repairable subscapularis**).
  - **Subacromial balloon spacer** (InSpace: **no benefit over debridement in START:REACTS trial**).
  - **Tendon transfers:**
  - **Posterosuperior tears (loss of ER): latissimus dorsi transfer (Gerber)** (**contraindicated with subscapularis deficiency**), **lower trapezius transfer (Elhassan; with Achilles allograft)** (more in-phase with ER).
  - **Anterosuperior/subscapularis tears: pectoralis major transfer** (**sternal head passed under the clavicular head (behind the conjoint**

tendon: more anatomic vector)), **latissimus dorsi anterior transfer**.
  - **Reverse total shoulder arthroplasty** (**elderly, pseudoparalysis, cuff tear arthropathy**).
  - **Interposition grafts/patch augmentation**.

7. Related conditions

- **Calcific tendinitis: supraspinatus (commonest)**, **women aged 30-50**, **hydroxyapatite (calcium hydroxyapatite) deposition**; **not degenerative** (cell-mediated). **Uhthoff stages: precalcific (fibrocartilaginous metaplasia)**, **calcific (formative phase, resting phase, resorptive phase: most painful, acute; deposit fluffy/ill-defined)**, **postcalcific (reconstitution)**. **Gärtner classification** (type I well-defined dense; II mixed; III cloudy translucent: resorptive). **Treatment: NSAIDs, physio**, **ultrasound-guided barbotage (lavage)** (best evidence), **extracorporeal shock wave therapy (ESWT)**, **arthroscopic excision**.
- **Frozen shoulder (adhesive capsulitis): diabetes (strongest association; about 10-20% of diabetics; worse prognosis)**, thyroid disease, Dupuytren, Parkinson, post-surgery, **women aged 40-60**. **Loss of active and passive ER (with arm at side) is the hallmark**; normal radiographs. **Phases: freezing (painful), frozen (stiff), thawing** (**about 1-3 years**). **Pathology: fibroblast/myofibroblast proliferation** (**Dupuytren-like**, type III collagen) in the **rotator interval and CHL**. **Treatment: intra-articular steroid injection (best early evidence)**, physiotherapy, **hydrodilatation**, **manipulation under anaesthesia (MUA)**, **arthroscopic capsular release** (**UK FROST trial**: all three (early structured physio + steroid, MUA, ACR) improved similarly; ACR more costly/risky).
- **SLAP lesions (superior labrum anterior to posterior) (Snyder): I** degenerative fraying (debride); **II superior labrum and biceps anchor detached** (**commonest**: repair in young overhead athletes; **biceps tenodesis in over 35-40 years**); **III bucket-handle tear of the labrum, biceps intact** (excise); **IV** bucket-handle extending into the biceps (repair/tenodesis depending on biceps involvement). **Tests: O'Brien (active compression)**, **Speed's**, **dynamic labral shear (O'Driscoll)**. **Spinoglenoid paralabral cyst** (suprascapular nerve compression: infraspinatus weakness).

- **Subscapularis tears and LHB instability**: **medial subluxation of the LHB** suggests a subscapularis tear.
- **Os acromiale: failure of fusion of acromial ossification centres (meso-acromion commonest)**; **fuse by about 25 years**; can cause impingement; **ORIF with bone graft** if symptomatic.

Exam pearl: **Supraspinatus/infraspinatus = suprascapular nerve; teres minor = axillary; subscapularis = upper/lower subscapular. Belly-press/bear-hug: subscapularis; hornblower: teres minor; ER lag: infraspinatus.**

Exam pearl: **Goutallier 3-4 (fat equal/greater than muscle) = irreversible, high re-tear. AHI under 7 mm = massive tear. CSA over 35 degrees = cuff tear. Acromioplasty no better than placebo (CSAW). Re-tear risk: age, size, retraction, fatty infiltration, smoking.**

Exam pearl: **Massive irreparable tears: SCR (young, no OA), LD or lower trapezius transfer for posterosuperior; pectoralis major transfer for subscapularis; RTSA for elderly pseudoparalysis/CTA. Preserve the CA arch (anterosuperior escape). Frozen shoulder: diabetes; loss of passive ER; FROST trial. SLAP II commonest (repair young; tenodesis older).**

Shoulder instability

1. Anatomy and stabilisers

- **Glenohumeral joint:** the most mobile and most commonly dislocated large joint. **Glenoid surface area is about one-third to one-quarter of the humeral head** ("golf ball on a tee"); glenoid **retroverted about 5-7 degrees**, superiorly tilted about 5 degrees; humeral head retroverted about 30 degrees.
- **Static stabilisers:**
  - **Labrum** (deepens the glenoid by about **50%**, increases contact area, anchors the IGHL; **sublabral foramen** and **Buford complex (absent anterosuperior labrum + cord-like MGHL)** are **normal variants**: do not repair).
  - **Glenohumeral ligaments:**
  - **SGHL:** restraint to **inferior translation in adduction** (with the coracohumeral ligament; **rotator interval**),
  - **MGHL: anterior translation at about 45 degrees abduction** with external rotation (absent in about 30%),
  - **IGHL complex (anterior band, axillary pouch, posterior band): the most important**: **anterior band** resists **anterior translation in 90 degrees abduction and external rotation** (the **apprehension position**); **posterior band** resists **posterior translation in flexion and internal rotation**.
  - **Coracohumeral ligament**: inferior translation in adduction, external rotation (tight in frozen shoulder).
  - **Negative intra-articular pressure**, **concavity-compression**, **adhesion-cohesion**.
- **Dynamic stabilisers: rotator cuff (concavity-compression: most important dynamic)**, **long head of biceps**, **scapular stabilisers** (scapulothoracic rhythm about **2:1 glenohumeral:scapulothoracic**).
- **Rotator interval:** triangle between **supraspinatus and subscapularis** (base: coracoid): contains **SGHL, CHL, long head of biceps**: **interval closure** reduces inferior/posterior laxity (MDI).

2. Classification of instability

- **TUBS** (**T**raumatic, **U**nidirectional, **B**ankart lesion, **S**urgery) vs **AMBRI** (**A**traumatic, **M**ultidirectional, **B**ilateral, **R**ehabilitation, **I**nferior capsular shift if surgery). **Stanmore triangle (polar types):**
  - **Polar I:** traumatic structural,
  - **Polar II:** atraumatic structural (capsular laxity/dysplasia),
  - **Polar III: muscle patterning (abnormal muscle recruitment)**, non-structural: **physiotherapy/biofeedback; avoid surgery**.
- **Direction: anterior (about 95%)**, posterior (**about 2-5%: seizures, electrocution, posterior force with adduction/internal rotation**), inferior (**luxatio erecta**: arm fixed overhead; **axillary nerve and artery injuries**), multidirectional.

3. Traumatic anterior dislocation

- **Mechanism: abduction + external rotation + extension force**.
- **Associated lesions:**
  - **Bankart lesion** (**anteroinferior labrum + IGHL avulsion from the glenoid**): **in about 90% of young patients**. **Bony Bankart** (glenoid rim fracture).
  - **Hill-Sachs lesion** (**posterosuperolateral humeral head impaction** on the glenoid rim: **in about 80-90%**).
  - **ALPSA** (**anterior labroligamentous periosteal sleeve avulsion**: labrum displaced **medially and inferiorly** on the scapular neck: heals in a malposition: **higher recurrence**; mobilise before repair).
  - **Perthes lesion** (labrum avulsed with **intact periosteum**: non-displaced).
  - **GLAD** (**glenolabral articular disruption**: labral tear + articular cartilage injury; usually stable).
  - **HAGL** (**humeral avulsion of the glenohumeral ligament**: **"J sign"** on MRI (IGHL becomes J-shaped); **recurrent instability if missed**; **open or arthroscopic repair**).
  - **Greater tuberosity fracture** (**commoner in over 40s**; reduces with the shoulder in most cases; **displacement over 5 mm (over 3 mm in overhead athletes): fix**).

  - **Rotator cuff tears** (**over 40 years: about 30%; over 60 years: about 80%**: **"terrible triad" of the shoulder**: dislocation + cuff tear + nerve injury). **Persistent weakness after reduction in an older patient = cuff tear (until proved otherwise)**: ultrasound/MRI; **early repair**.
  - **Axillary nerve injury** (**about 5-35%, mostly neurapraxia**; **older patients**): check **regimental badge sensation** and deltoid **before and after reduction**.
  - **Axillary artery injury** (elderly with atherosclerosis).
- **Reduction techniques: Kocher** (traction, external rotation, adduction, internal rotation: **risk of fracture/nerve injury**), **Hippocratic**, **Stimson** (prone, hanging weight), **Milch** (abduction + external rotation), **FARES**, **Cunningham (massage technique)**, **scapular manipulation**; **post-reduction radiographs (AP + axillary/Velpeau)**.
- **Recurrence (the most important determinant is age at first dislocation):**
  - **Under 20 years: about 70-90%** (up to about **90%** in **contact athletes**, male),
  - **20-40 years: about 50-60%**,
  - **Over 40 years: about 10-15%** (cuff tears, stiffness more common than recurrence).
- **Immobilisation in external rotation** (to approximate the Bankart lesion) has **not consistently reduced recurrence** vs internal rotation sling.
- **Primary arthroscopic stabilisation after first dislocation** in **young (under about 25) contact/collision athletes** reduces recurrence (**about 80% recurrence non-operatively vs about 10-15% after surgery**); offered selectively.

4. Bone loss: the key to surgical choice

Glenoid bone loss

- **Inverted pear glenoid** (arthroscopic appearance: loss of anteroinferior bone, **over about 25% loss**).
- **Measurement: 3D CT (en face view of the glenoid)**: **best-fit inferior circle method**, **Pico method**, **glenoid width loss as percentage**.

- **Critical bone loss: historically about 20-25%**; **"subcritical" bone loss of about 13.5-17%** still raises recurrence and worsens outcomes after arthroscopic Bankart repair (**Shaha**).

Humeral bone loss and the glenoid track

- **Glenoid track concept (Yamamoto, Di Giacomo):** the zone of contact of the glenoid on the humeral head in **abduction/external rotation**: **glenoid track width = 0.83 x D (intact glenoid width) - d (glenoid bone loss width)**.
  - **Hill-Sachs interval (HSI) = width of the Hill-Sachs lesion + bone bridge** (to the cuff insertion).
  - **HSI over the glenoid track = "off-track" (engaging) Hill-Sachs** (it can **lever over the anterior glenoid rim** in the functional range): **high recurrence after isolated Bankart repair**.
  - **HSI under the glenoid track = "on-track"**.
- **Engaging Hill-Sachs (Burkhart):** lesion aligned parallel to the anterior glenoid in a functional position (abduction/external rotation): engages.

Treatment algorithm (Di Giacomo et al.)

Glenoid bone loss	Hill-Sachs	Treatment
Under 25% (or under subcritical threshold)	**On-track**	**Arthroscopic Bankart repair**
Under 25%	**Off-track**	**Arthroscopic Bankart + remplissage**
Over 25%	On-track	**Latarjet (bone block)**
Over 25%	Off-track	**Latarjet ± humeral procedure (remplissage/bone graft)** depending on track after Latarjet

- **ISIS score (Instability Severity Index Score; Balg and Boileau): over 6 points = high recurrence after arthroscopic Bankart (about 70%): Latarjet**:
  - **Age under 20 at surgery: 2 points**,

- **Competitive sport: 2 points**,
- **Contact or forced overhead sport: 1 point**,
- **Shoulder hyperlaxity (external rotation over 85 degrees or Gagey hyperabduction over 90 degrees): 1 point**,
- **Hill-Sachs visible on AP radiograph in external rotation: 2 points**,
- **Loss of glenoid contour on AP radiograph: 2 points**.

5. Surgical procedures

Arthroscopic Bankart repair

- **Suture anchors (at least 3 anchors, at least 4 mm apart, placed on the articular margin of the glenoid face (not medially on the neck))**, **inferior capsular shift** (plication), **anchors at the 3-5 o'clock (right shoulder)** region; **beach chair or lateral decubitus**.
- **Recurrence about 5-15% (higher with bone loss, young contact athletes, ISIS over 6, fewer than 3 anchors, ALPSA)**.
- **Open Bankart** (subscapularis split/takedown): historically the gold standard; **contact athletes** in some centres; **external rotation loss**.

Remplissage ("to fill")

- **Arthroscopic capsulotenodesis of the infraspinatus tendon (and posterior capsule) into the Hill-Sachs defect**: converts the intra-articular defect to extra-articular. **Small loss of external rotation**; reduces recurrence for off-track lesions.

Latarjet (coracoid transfer)

- **Coracoid process (about 2-3 cm) with the conjoint tendon** transferred to the **anteroinferior glenoid (flush with the glenoid face, 3-5 o'clock, below the equator)** through a **subscapularis split**, fixed with **two screws**.
- **"Triple blocking effect" (Patte):**
  - **Bony effect** (extends the glenoid arc),
  - **Sling effect of the conjoint tendon** (**most important in abduction-external rotation**: dynamic sling on the lower subscapularis and capsule),

    - **Capsular repair** (to the coracoacromial ligament stump: **Bankart effect**).
- **Bristow** = tip of the coracoid only, placed vertically, fixed with one screw (less bone).
- **Complications: graft malposition (too lateral: OA; too medial: recurrence)**, **graft non-union/osteolysis/fracture**, **screw problems**, **musculocutaneous nerve injury** (enters the coracobrachialis about **5 cm below the coracoid**: beware traction), **axillary nerve injury**, **loss of external rotation**, **glenohumeral OA (long-term)**, recurrence (low, about 2-5%), infection, haematoma.
- **Free bone block (iliac crest (Eden-Hybinette), distal tibial allograft)** for **massive bone loss or failed Latarjet**.

Other

- **HAGL repair**, **humeral head allograft/disimpaction** for large Hill-Sachs, **humeral rotational osteotomy (historical: Weber)**.
- **Thermal capsulorrhaphy: abandoned** (capsular necrosis, **chondrolysis**, axillary nerve injury). **Intra-articular bupivacaine pain pumps**: **glenohumeral chondrolysis**.

6. Posterior instability

- **Traumatic posterior dislocation: seizures, electrocution, ECT**: **often missed (about 50-80% missed initially)**; **arm locked in internal rotation**, **cannot externally rotate**; **AP radiograph: "light bulb sign"**, loss of the half-moon overlap, **rim sign (over 6 mm)**; **axillary view/CT essential**. **Reverse Hill-Sachs lesion (McLaughlin lesion)** of the **anteromedial humeral head**.
    - **Treatment by defect size: under 25%**: closed reduction, immobilise in external rotation; **25-50%**: **McLaughlin procedure (subscapularis tendon transfer into the defect)** or **modified McLaughlin (lesser tuberosity transfer: Neer)**, **allograft reconstruction**; **over 50%** or chronic over 6 months: **hemiarthroplasty/arthroplasty**.
- **Recurrent posterior subluxation: athletes (football linemen, weightlifters, swimmers)**: **reverse Bankart lesion**, **Kim lesion** (incomplete, concealed avulsion of the posteroinferior labrum); **Jerk test,**

Kim test; **posterior labral repair/capsular shift**; **posterior bone block** for glenoid bone loss; **glenoid opening-wedge osteotomy** for **excess glenoid retroversion** (over about 15-20 degrees).

7. Multidirectional instability (MDI)

- **Inferior + anterior and/or posterior instability**; **generalised hyperlaxity (Beighton score)**, connective tissue disorders (**Ehlers-Danlos**), **patulous inferior capsule**, enlarged rotator interval; often **bilateral, atraumatic**.
- **Sulcus sign (over 2 cm, persistent in external rotation = rotator interval incompetence)**, **Gagey hyperabduction test** (**over 105 degrees = IGHL laxity**).
- **Treatment: prolonged rehabilitation (at least 6 months)**: rotator cuff and scapular stabiliser strengthening (**most succeed**); **surgery** for failure: **open inferior capsular shift (Neer)** or **arthroscopic capsular plication ± rotator interval closure**. **Avoid surgery in voluntary/muscle-patterning instability (Polar III)** and **EDS** (high failure).

8. Examination tests

- **Anterior: apprehension test** (abduction-external rotation), **relocation test (Jobe: posterior force relieves apprehension)**, **release (surprise) test** (**most specific**), **anterior drawer**, **load and shift**.
- **Posterior: Jerk test**, **Kim test**, posterior drawer, posterior load and shift.
- **Inferior/MDI: sulcus sign**, **Gagey**.
- **Laxity ≠ instability:** instability = symptomatic pathological translation.

> Exam pearl: **Age at first dislocation is the strongest predictor of recurrence (under 20: up to 90%). Over 40: think cuff tear and axillary nerve. IGHL anterior band = main restraint in abduction-external rotation.**

Exam pearl: **Glenoid track = 0.83D - d; HSI over the track = off-track (engaging) = add remplissage. Glenoid loss over 20-25% (subcritical 13.5-17%) = Latarjet. ISIS over 6 = Latarjet. Latarjet: sling effect of the conjoint tendon is the main mechanism; musculocutaneous nerve at risk.**

Exam pearl: **Posterior dislocation: seizure/electrocution, locked internal rotation, light bulb sign, reverse Hill-Sachs; McLaughlin transfer for 25-50% defects. HAGL = J sign. ALPSA heals medially (higher recurrence). MDI: rehab first; inferior capsular shift; avoid surgery in muscle patterning.**

Elbow in throwers

1. Throwing biomechanics

- **Phases of throwing: wind-up**, **early cocking**, **late cocking** (maximum shoulder abduction and **external rotation**), **acceleration** (**most rapid motion in sport: shoulder internal rotation up to about 7,000 degrees/second**), **deceleration** (eccentric posterior cuff load), **follow-through**.
- **Elbow valgus load peaks in late cocking and early acceleration**: about **64 Nm of valgus torque** at the elbow, which **exceeds the ultimate tensile strength of the UCL (about 32-34 Nm)**: the remainder is resisted by the **flexor-pronator mass, the radiocapitellar joint and the posteromedial olecranon**.
- **Valgus extension overload (VEO):** valgus + rapid extension creates:
  - **Medial tension**: **UCL injury**, **medial epicondyle apophysitis/avulsion**, **flexor-pronator strain**, **ulnar neuritis**.
  - **Lateral compression**: **radiocapitellar overload: OCD of the capitellum**, **Panner disease**, radial head changes, loose bodies.
  - **Posterior shear (posteromedial impingement)**: **olecranon osteophytes (posteromedial)**, **olecranon stress fractures**, loose bodies.
- **Kinetic chain: legs and trunk generate about 50% of velocity/force**; **scapular dyskinesis**, **GIRD** (glenohumeral internal rotation deficit over about 20 degrees or total arc loss), and poor mechanics increase elbow and shoulder loads.

2. Ulnar collateral ligament (UCL / medial collateral ligament)

- **Three bundles:**
  - **Anterior bundle (the most important: primary restraint to valgus from about 20-120 degrees)**: from the **anteroinferior surface of the medial epicondyle** to the **sublime tubercle** of the coronoid; **anterior band tight in extension, posterior band tight in flexion**.
  - **Posterior bundle** (floor of the cubital tunnel; tight in flexion: **contracture causes loss of flexion**).

- **Transverse bundle (Cooper ligament)**: no stabilising role.
- **Valgus stability: in full extension: the anterior capsule, UCL and bony articulation each contribute about one-third**; **at 90 degrees: the UCL (anterior bundle) provides about 55-70%**; **radial head = secondary valgus stabiliser** (important if UCL deficient).

Injury

- **Chronic attenuation** (repetitive microtrauma) or **acute rupture** ("pop" during throwing); **medial elbow pain in late cocking/early acceleration**, **loss of velocity and control**, **ulnar nerve paraesthesia** (about 40%).
- **Examination tests:**
  - **Moving valgus stress test (O'Driscoll)**: **most sensitive (about 100%)**: valgus stress while rapidly extending from full flexion: **pain between about 70 and 120 degrees ("shear angle")**,
  - **Milking manoeuvre** (pull on the thumb with shoulder abducted, elbow flexed at 90 degrees),
  - **Valgus stress test at about 30 degrees of flexion** (unlocks the olecranon).
- **Imaging: stress radiographs** (medial joint opening over about 3 mm side-to-side difference), **MR arthrography** (**"T sign"**: **undersurface (partial) tear at the sublime tubercle** with dye tracking distally along the ulna), **dynamic ultrasound**.
- **Treatment:**
  - **Non-operative:** rest from throwing (about 6-12 weeks), **rehabilitation (flexor-pronator strengthening, kinetic chain)**, **PRP (partial tears)**; good for **partial tears and non-throwers**.
  - **Reconstruction ("Tommy John" surgery; Jobe 1974):**
  - **Graft: palmaris longus (ipsilateral; absent in about 15%)**, **gracilis**, plantaris.
  - **Original Jobe technique: flexor-pronator detachment + submuscular ulnar nerve transposition (ulnar neuropathy in about 20%+)**.
  - **Modified Jobe** (**muscle-splitting approach**, no routine transposition).
  - **Docking technique (Altchek)**: **single humeral tunnel with the graft ends docked**; **higher return-to-play rates**.

  - **DANE TJ** (hybrid).
  - **Return to play about 80-90%**, after about **12-18 months** (pitchers).
  - **UCL repair with internal brace (suture tape augmentation)**: **proximal or distal avulsions with good tissue quality in young athletes**; **faster return (about 6-7 months)**.
- **Complications: ulnar nerve neuritis (commonest)**, medial antebrachial cutaneous nerve injury, **medial epicondyle fracture (tunnel)**, graft failure, stiffness.

3. Valgus extension overload and posteromedial impingement

- **Posteromedial olecranon osteophytes** impinge on the **olecranon fossa** in extension: **posterior elbow pain at ball release**, **loss of terminal extension**, **loose bodies**, **chondromalacia of the posteromedial trochlea ("kissing lesion")**.
- **Test: valgus extension overload test** (repeated forced extension with valgus: posteromedial pain).
- **Treatment:** rehab, **arthroscopic osteophyte excision** (**excise only the osteophyte, avoid over-resection of the normal olecranon: increases UCL strain**); **always assess the UCL** (VEO and UCL insufficiency coexist).
- **Olecranon stress fractures** (**transverse or oblique**; **triceps traction** or impingement); **ORIF with compression screw** if not healing.

4. Lateral compression: capitellum lesions

	Panner disease	**Osteochondritis dissecans (OCD) of the capitellum**
Age	**Under 10 years (about 5-10)**	**Adolescents (about 12-16)**
Pathology	**Osteochondrosis**: **fragmentation of the whole ossific nucleus of the capitellum**	**Focal subchondral lesion**, anterolateral capitellum; **articular cartilage involved**
Prognosis	**Self-limiting**: reossifies	**Variable**: loose bodies, arthritis

	Panner disease	Osteochondritis dissecans (OCD) of the capitellum
Treatment	**Rest, avoid throwing, symptomatic**; no surgery	**Stable** (open capitellar physis, good ROM, intact cartilage): **rest 3-6 months**; **unstable/failed**: **arthroscopic debridement + microfracture** (small lesions), **fixation**, **osteochondral autograft (OATS from the knee)** for large lesions (**over about 50% of capitellar width or involving the lateral margin**)

- **Typical patients: baseball pitchers, gymnasts** (upper limb weight bearing).
- **Pseudodefect of the capitellum** (normal posterior non-articular groove: do not mistake for OCD).

5. Medial epicondyle problems in skeletally immature throwers

- **"Little League elbow"** (**medial epicondyle apophysitis**): **medial elbow pain**, **widening/fragmentation of the medial epicondyle apophysis**; **rest from throwing (about 4-6 weeks)** and gradual return with **pitch count limits** (**pitch-count guidelines by age**, avoid pitching year-round, avoid curveballs in young pitchers (controversial), no pitching with fatigue).
- **Medial epicondyle avulsion fracture: displacement over 5 mm (or 2 mm in throwers/dominant arm)**, **incarceration in the joint (absolute indication)**, **ulnar nerve dysfunction**, **valgus instability**: **ORIF (screw)**.

6. Other elbow conditions in athletes

- **Lateral epicondylitis (tennis elbow): ECRB origin (angiofibroblastic tendinosis: Nirschl)**; **ages 35-55**; **Cozen test (resisted wrist extension)**, **Mill test**, **Maudsley test (resisted middle finger extension)**. **Self-limiting (about 80-90% resolve within 1 year)**; **physiotherapy (eccentric loading)**, **counterforce brace**; **steroid injection: short-term benefit but worse outcomes at 1 year than wait-and-see (Bisset, Coombes)**; **PRP** (mixed evidence); **surgery (open, percutaneous or arthroscopic ECRB release/debridement)** after **6-12 months** of failed conservative care. **Rule out radial tunnel syndrome and posterolateral rotatory instability**

(PLRI) (**iatrogenic LUCL injury during lateral release/debridement**: stay anterior to the radial head equator).

- **Medial epicondylitis (golfer's elbow): pronator teres and FCR origin**; **ulnar neuritis coexists in about 20-60%**; similar treatment.
- **Distal biceps tendon rupture: eccentric load on a flexed elbow**, **dominant arm of men aged 40-60**, **smokers, anabolic steroids**; **hook test (O'Driscoll: most sensitive and specific)**, **reverse Popeye**, **loss of supination strength (about 40-50%) and flexion strength (about 30%)**. **Early repair (within 2-3 weeks)** via **single-incision (anchors, cortical button: higher LABCN (lateral antebrachial cutaneous nerve) injury: commonest complication)** or **two-incision (Boyd-Anderson: heterotopic ossification/radioulnar synostosis)**. **Chronic (over about 6 weeks): reconstruction with graft** (Achilles allograft). **Lacertus fibrosus intact** may limit retraction.
- **Distal triceps rupture:** rare; **anabolic steroids, olecranon bursitis, renal failure, local steroid injection**; **"fleck sign" (olecranon avulsion)**; repair.
- **PLRI (posterolateral rotatory instability): LUCL (lateral ulnar collateral ligament) insufficiency**: **after elbow dislocation**, iatrogenic, cubitus varus; **lateral pivot-shift test (O'Driscoll: supination, valgus, axial load in extension: subluxation, reduces with flexion at about 40 degrees)** (best under anaesthesia), **chair push-up test**, **table-top relocation test**; **LUCL reconstruction (docking, palmaris longus)**.
- **Elbow arthroscopy portals:**
  - **Proximal anteromedial portal (about 2 cm proximal to the medial epicondyle, anterior to the intermuscular septum)**: **medial antebrachial cutaneous nerve**; **ulnar nerve** (posterior to the septum: **check for ulnar nerve subluxation/previous transposition before medial portals**),
  - **Anterolateral portal**: **radial nerve (PIN)** (the **closest major nerve: about 4-7 mm; distended joint and flexion move it away**),
  - **Proximal anterolateral portal** (safer from the radial nerve than distal anterolateral),
  - **Direct lateral ("soft spot") portal**: posterior antebrachial cutaneous nerve,
  - **Posterior (posterocentral, posterolateral)**.

    - **Most common complication of elbow arthroscopy: transient nerve palsy** (radial/ulnar/median).
- **Elbow stiffness (post-traumatic):** functional arc **30-130 degrees (Morrey)** (**modern tasks need about 27-149 degrees**); **arthroscopic or open (column procedure) release**; **ulnar nerve decompression if flexion under about 90-100 degrees pre-op** (risk of post-op ulnar neuritis when flexion improves).

> Exam pearl: **Valgus torque in late cocking/early acceleration exceeds the UCL: flexor-pronator, radiocapitellar joint and olecranon share load. VEO = medial tension (UCL), lateral compression (capitellar OCD/Panner), posterior shear (posteromedial osteophytes).**

> Exam pearl: **UCL anterior bundle = primary valgus restraint 20-120 degrees; moving valgus stress test most sensitive (pain 70-120 degrees); MR arthrography T sign. Reconstruction (palmaris longus, docking), ulnar neuritis = commonest complication. Internal-brace repair for avulsions in young throwers.**

> Exam pearl: **Panner (under 10, whole capitellum, self-limiting) vs capitellar OCD (adolescents, focal, may need surgery: OATS for large/lateral-margin lesions). Medial epicondyle fracture: incarcerated fragment = absolute surgery; throwers fix over 2 mm. Distal biceps: hook test; single-incision = LABCN injury; two-incision = synostosis.**

Hip arthroscopy and FAI

1. Femoroacetabular impingement (FAI): concept

- **Abnormal contact between the proximal femur and acetabular rim** during motion (especially **flexion + internal rotation**): **labral tears and chondral damage**: a **cause of early OA** in young adults (Ganz).
- **Warwick Agreement (2016) definition of FAI syndrome: triad of symptoms (motion- or position-related hip/groin pain), clinical signs (positive impingement test, restricted internal rotation) and imaging findings (cam or pincer morphology)**. **Morphology alone is not FAI** (common in asymptomatic athletes).

2. Types

	Cam	Pincer
Pathology	**Femoral**: **aspherical femoral head-neck junction** (reduced offset, "pistol-grip" deformity); jammed into the acetabulum in flexion/IR	**Acetabular**: **global or focal over-coverage** (retroversion, coxa profunda, protrusio)
Typical patient	**Young, athletic males** (adolescent sports during physeal closure: **repetitive loading of the developing physis**); **SCFE sequelae**, Perthes	**Middle-aged active women**

	Cam	Pincer
Damage pattern	**"Outside-in" abrasion**: **chondrolabral separation**, **delamination of the anterosuperior acetabular cartilage** (**carpet lesion/wave sign**): **more cartilage damage**	**Labral crushing/degeneration first** (linear contact); **contrecoup lesion posteroinferiorly** (head levered out); ossification of the labrum
Measures	**Alpha angle (over 55 degrees)** (Notzli; on 45-degree Dunn view or radial MRI), **head-neck offset ratio (under 0.17)**, **pistol-grip deformity**, **anterior offset**	**Crossover sign** (retroversion: anterior wall crosses the posterior wall), **posterior wall sign** (posterior wall medial to the head centre), **ischial spine sign** (projection of the ischial spine into the pelvis: retroversion), **lateral centre-edge angle (LCEA) over 40 degrees**, **coxa profunda (acetabular fossa medial to the ilioischial line)**, **protrusio (femoral head medial to ilioischial line)**, **Tönnis angle (acetabular index) under 0 degrees**

- **Mixed (combined) FAI: commonest (about 70-85%)**.
- **Subspine (AIIS) impingement:** prominent **anterior inferior iliac spine** (Hetsroni types I-III) (**healed rectus femoris avulsion**): **AIIS decompression**.
- **Ischiofemoral impingement:** narrowing between the ischium and lesser trochanter (**quadratus femoris oedema** on MRI).
- **Femoral version abnormalities: femoral retroversion** worsens impingement; **excessive femoral anteversion/acetabular dysplasia**: **instability, not impingement**: hip arthroscopy alone may worsen symptoms.

3. Clinical evaluation

- **Groin pain ("C sign": hand cupped over the lateral hip)**, worse with **sitting, flexion, pivoting**, mechanical symptoms (catching, clicking), **reduced internal rotation in flexion**.

- **Tests:**
  - **FADIR (anterior impingement test: flexion, adduction, internal rotation)**: **highly sensitive, low specificity**,
  - **FABER (Patrick)**: posterior/SI or posterior impingement,
  - **Posterior impingement test** (extension + external rotation),
  - **Log roll** (instability/intra-articular),
  - **Dial test** (capsular laxity),
  - **Stinchfield (resisted straight leg raise)**: intra-articular pathology.
- **Diagnostic intra-articular local anaesthetic injection**: pain relief confirms an intra-articular source.

4. Imaging

- **AP pelvis (standardised: coccyx tip about 1-3 cm above the symphysis, centred)**: **crossover sign, posterior wall sign, ischial spine sign, LCEA, Tönnis angle, joint space (Tönnis OA grade)**, **pistol-grip deformity**.
- **45-degree Dunn view / frog-leg lateral / cross-table lateral**: **alpha angle, head-neck offset**.
- **False profile view (Lequesne)**: **anterior centre-edge angle (ACEA under 20 degrees = anterior undercoverage)**, AIIS.
- **MR arthrography (gadolinium) or 3T MRI** (labral tears: **anterosuperior** commonest, cartilage damage), **CT with 3D reconstruction** (cam/pincer mapping, **femoral version**, AIIS).

Acetabular dysplasia and instability measures (important to exclude)

- **LCEA (Wiberg): normal 25-40 degrees**; **under 20 degrees = dysplasia**; **20-25 degrees = borderline dysplasia**.
- **Tönnis (acetabular) angle: normal 0-10 degrees**; **over 10 degrees = dysplasia (steep roof)**; **under 0 degrees = over-coverage (pincer)**.
- **Femoral head extrusion index** (over 25% = uncovered), **Shenton line break**, **FEAR index** (femoro-epiphyseal acetabular roof index: helps decide instability in borderline dysplasia).
- **Dysplasia (LCEA under 20) treatment = periacetabular osteotomy (PAO; Bernese/Ganz)**: **isolated arthroscopic labral debridement/capsulotomy in dysplasia can cause catastrophic instability**.

5. Indications and contraindications for FAI surgery

- **Indications: symptomatic FAI syndrome** failing non-operative management (**activity modification, physiotherapy (core and hip strengthening), NSAIDs, injection**) for at least about 3 months, with **preserved joint space**.
- **Contraindications / poor prognostic factors: established OA (Tönnis grade 2 or more, joint space under 2 mm)**: **poor outcomes; conversion to THA**, **older age (over 40-50)**, **significant dysplasia (LCEA under 20)** (unless combined with PAO), **severe chondral damage (Outerbridge IV)**, **high BMI**, workers' compensation.
- **Evidence: UK FASHIoN trial** (and **Australian FAIT**): **hip arthroscopy improved patient-reported outcomes more than best conservative care (physiotherapy)** at 12 months in FAI syndrome.

6. Surgical technique

- **Supine or lateral position on a traction table**, **well-padded perineal post** (or **post-less**), **traction (about 8-10 mm of joint distraction)**; **limit traction time to under about 2 hours** (**pudendal nerve neurapraxia** (perineal post), **sciatic nerve**, **lateral femoral cutaneous nerve** (anterior portal), **perineal skin injury**, **scrotal/labial injury**).
- **Portals:**
  - **Anterolateral (first; under fluoroscopy; 1 cm anterior and 1 cm superior to the greater trochanter tip)**: **superior gluteal nerve** (about 4-6 cm superior),
  - **Mid-anterior / anterior**: **lateral femoral cutaneous nerve (closest structure: about 3 mm)**, **femoral neurovascular bundle** (medial: avoid crossing the line from the ASIS downward),
  - **Posterolateral**: **sciatic nerve** (keep the leg in neutral/internal rotation).
- **Central compartment** (with traction): labrum, chondral surfaces, ligamentum teres, acetabular fossa; **peripheral compartment** (without traction, hip flexed): **head-neck junction (cam)**, synovium, loose bodies.
- **Interportal or T-capsulotomy**; **capsular repair/closure** (**reduces post-op instability**, especially in **borderline dysplasia, hyperlaxity, women**).

- **Pincer: acetabular rim trimming (acetabuloplasty)** with **labral takedown and re-fixation** (**avoid over-resection: iatrogenic instability/dysplasia**; LCEA should remain over 25 degrees).
- **Labrum: repair/refixation (suture anchors) preferred over debridement** (better outcomes; **labral seal effect**: maintains intra-articular fluid pressure, cartilage nutrition, stability); **labral reconstruction (allograft/ITB autograft)** for irreparable/deficient labra in young patients.
- **Cam: femoroplasty (osteochondroplasty)** of the head-neck junction under fluoroscopy and dynamic assessment; **avoid over-resection (over about 30% of the neck diameter: femoral neck fracture risk)**; **protect the retinacular (lateral epiphyseal) vessels** (**posterosuperior neck: branches of the medial femoral circumflex artery**: **avoid resection at the 12 o'clock-posterior region**: **AVN**).
- **Chondral lesions**: **microfracture** for small full-thickness acetabular lesions.
- **Open surgical hip dislocation (Ganz: trochanteric flip osteotomy, preserving the MFCA by staying anterior to the short external rotators; the obturator externus protects the deep branch)**: for **complex/global deformities (severe pincer, protrusio, posterior cam, Perthes deformity, SCFE)**.
- **Periacetabular osteotomy (PAO)** for dysplasia (± arthroscopy); **femoral derotation osteotomy** for abnormal version.

7. Complications of hip arthroscopy

- **Neurapraxia** (**pudendal (perineal post: commonest neurological), sciatic, femoral, LFCN**), **traction injuries**, **iatrogenic chondral/labral damage** (scuffing on portal entry), **instrument breakage**.
- **Heterotopic ossification** (**NSAID prophylaxis (naproxen) after femoroplasty**).
- **Iatrogenic instability/dislocation** (unrepaired capsulotomy, over-resection, dysplasia, hyperlaxity).
- **Femoral neck fracture** (over-resection), **AVN** (rare).
- **Abdominal fluid extravasation** (**intra-abdominal compartment syndrome**; risk with iliopsoas release, high pump pressure, prolonged surgery).

- **DVT**, **infection** (rare), **under-resection** (**residual impingement: commonest cause of revision**), **adhesions**.

8. Other hip arthroscopy indications and extra-articular conditions

- **Ligamentum teres tears** (instability, dysplasia), **loose bodies** (synovial chondromatosis, trauma), **septic arthritis** (washout), **pigmented villonodular synovitis (diffuse TGCT)**, **gluteus medius/minimus tears ("rotator cuff of the hip")**: **endoscopic or open repair**; **greater trochanteric pain syndrome** (gluteal tendinopathy: **education + loading exercise (LEAP trial) best long-term**; ESWT; steroid injection short-term).
- **Snapping hip:**
  - **External (ITB over the greater trochanter)**: **visible snap**; ITB release/Z-plasty,
  - **Internal (iliopsoas tendon over the iliopectineal eminence or femoral head)**: **audible snap**, **extension from a flexed-abducted-externally rotated position**; **iliopsoas release (at the lesser trochanter or transcapsular)** if refractory (weakness of hip flexion),
  - **Intra-articular** (labral tears, loose bodies).
- **Deep gluteal syndrome** (sciatic nerve entrapment: **piriformis**, fibrous bands): endoscopic decompression.
- **Athletic pubalgia ("sports hernia"/core muscle injury)** and **adductor-related groin pain** (**Doha agreement** classification: adductor-, iliopsoas-, inguinal-, pubic-related groin pain); **osteitis pubis**.
- **Femoral neck stress fractures: compression side (inferomedial: protected weight bearing; fix if fracture line over 50% of the neck width)** vs **tension side (superolateral: urgent internal fixation)** (**female athlete triad/RED-S**).

> Exam pearl: **Cam (young men, alpha angle over 55, outside-in chondral delamination) vs pincer (middle-aged women, crossover/posterior wall/ischial spine signs, LCEA over 40, contrecoup). Mixed commonest. FADIR sensitive.**

Exam pearl: **Arthroscopy contraindicated with joint space under 2 mm/Tönnis 2+ and in dysplasia (LCEA under 20: needs PAO). Repair the labrum (seal) and the capsule. Avoid over-resection (neck fracture) and posterosuperior retinacular vessels (AVN). Under-resection = commonest cause of revision. FASHIoN: arthroscopy beats physio.**

Exam pearl: **Portals: anterior portal = LFCN closest; anterolateral = superior gluteal nerve; posterolateral = sciatic nerve. Perineal post = pudendal neurapraxia. HO prophylaxis with NSAIDs. Fluid extravasation into the abdomen = abdominal compartment syndrome.**

Sports medicine and anti-doping

1. Muscle injury and healing

- **Muscle strain:** commonest at the **myotendinous junction**, in **two-joint muscles** with **eccentric contraction** (**hamstrings (biceps femoris long head commonest)**, rectus femoris, medial gastrocnemius ("tennis leg"), adductor longus).
- **Healing phases: destruction/inflammation (days 1-3)**, **repair (regeneration of myofibres from satellite cells + scar formation, days 3-21)**, **remodelling** (weeks-months). **Satellite cells** = muscle stem cells.
- **Grading: grade I** (stretch, under 5% fibres), **II** (partial tear), **III** (complete rupture); **British Athletics Muscle Injury Classification (BAMIC)**: **0-4** with **a (myofascial), b (musculotendinous junction), c (intratendinous: longer return to play, higher recurrence)**.
- **Hamstring injuries:**
  - **Proximal hamstring avulsion** (**forced hip flexion with knee extension: water skiing**): **surgical repair** for **complete avulsion of 3 tendons**, or **2 tendons with over 2 cm retraction** in active patients; **early repair (within about 4-6 weeks)** gives better outcomes; **sciatic nerve** proximity.
  - **Prevention: Nordic hamstring eccentric programme** (reduces injuries by about 50%).
- **Treatment of muscle strains: PEACE & LOVE** (protection, elevation, avoid anti-inflammatories early, compression, education; load, optimism, vascularisation, exercise) has largely replaced **RICE/PRICE**; early controlled mobilisation; avoid immobilisation over a few days.
- **Delayed-onset muscle soreness (DOMS): 24-72 hours after unaccustomed eccentric exercise**; microtrauma; **raised CK**; self-limiting.
- **Myositis ossificans** (after contusion, **quadriceps**): **zonal pattern (mature peripheral ossification, immature centre)** (vs osteosarcoma: **reverse zoning**); **avoid massage/aggressive stretching early**; excise only after maturation (**about 6-12 months, bone scan cold**) if symptomatic.
- **Exertional rhabdomyolysis: CK over 5 times normal (often over 10,000)**, **myoglobinuria (dark urine)**, **acute kidney injury**; **sickle cell trait**, heat,

dehydration; **aggressive IV fluids**.

2. Tendon injury and tendinopathy

- **Tendon structure: type I collagen (about 95%)**, **tenocytes**, proteoglycans (**decorin**); hierarchy: fibril, fibre, fascicle, tendon; **endotenon/epitenon/paratenon (vascular, in tendons without sheaths)**.
- **Tendon healing:** inflammatory (days), proliferative (**type III collagen**, weeks), remodelling (**type III replaced by type I**, months); **never reaches original strength (about 70-80%)**.
- **Tendinopathy: failed healing response** (**not inflammation: "tendinosis"**): **disorganised collagen, increased ground substance, neovascularisation with nerve ingrowth (pain)**. **Continuum model (Cook and Purdam): reactive tendinopathy, tendon disrepair, degenerative tendinopathy**.
- **Treatment: load management, eccentric (Alfredson) or heavy slow resistance programmes (best evidence)**; **isometrics for pain relief**; **ESWT**; **PRP** (inconsistent); **steroid injections around load-bearing tendons (Achilles, patellar): risk of rupture: avoid intratendinous**.
- **Fluoroquinolones** (ciprofloxacin): **tendinopathy/rupture (Achilles)**, especially **older patients, with corticosteroids, renal impairment**. **Statins** (possible).

3. Bone stress injuries

- **Stress fractures** (**fatigue fractures in normal bone** from repetitive loading) vs **insufficiency fractures** (normal load on abnormal bone).
- **Risk factors: sudden increase in training**, **female athlete triad/RED-S**, **low vitamin D**, **low BMI**, **military recruits**, **cavus feet**, leg-length discrepancy, hard surfaces, poor footwear.
- **Imaging:** radiographs often normal initially (**periosteal reaction after 2-3 weeks**); **MRI (most sensitive and specific; Fredericson grading 1-4)**; bone scan (sensitive, non-specific).
- **High-risk stress fractures (tension side, poor blood supply: prone to non-union/displacement: often surgery or strict non-weight-bearing):**
  - **Femoral neck (tension: superolateral)**,

  - **Anterior cortex of the mid-tibia ("dreaded black line")** (**jumpers**; IM nailing if not healing),
  - **Medial malleolus**,
  - **Talus (lateral process)**,
  - **Navicular (central third: avascular)**: **CT**; **non-weight-bearing cast 6-8 weeks** or **screw fixation** in athletes,
  - **Proximal fifth metatarsal (Jones/zone 2-3: metaphyseal-diaphyseal junction)**: **intramedullary screw** in athletes,
  - **Sesamoids of the hallux**,
  - **Patella**,
  - **Base of the second metatarsal (ballet dancers)**.
- **Low-risk (compression side): posteromedial tibia (commonest stress fracture overall)**, **metatarsal shafts (2nd and 3rd: "march fracture")**, femoral shaft, **fibula**, calcaneus, **femoral neck compression side (inferomedial)**: **relative rest, modified activity**.
- **Medial tibial stress syndrome ("shin splints")**: **diffuse posteromedial tibial pain (distal two-thirds)**, **periostitis/traction of the soleus/FDL/tibialis posterior**; **MRI: periosteal oedema without fracture line**; rest, gradual return, **correct training errors**.
- **Chronic exertional compartment syndrome (CECS): anterior compartment commonest (then deep posterior)**; **pain with exercise, relieved by rest, ± numbness/foot drop**; **compartment pressure testing**: **(Pedowitz criteria) pre-exercise ≥15 mmHg, 1 minute post-exercise ≥30 mmHg, or 5 minutes post-exercise ≥20 mmHg**; **fasciotomy** (anterior and lateral; **superficial peroneal nerve** at risk where it pierces the fascia about **10-12 cm above the lateral malleolus**).
- **Popliteal artery entrapment syndrome** (young athletes: claudication; **medial head of gastrocnemius anomaly**), **external iliac artery endofibrosis** (cyclists): differentials of exertional leg pain.

4. Female athlete triad / RED-S

- **Female athlete triad (ACSM): low energy availability (with or without disordered eating), menstrual dysfunction (functional hypothalamic amenorrhoea), low bone mineral density** (**each on a spectrum**).

- **Relative Energy Deficiency in Sport (RED-S, IOC 2014; updated 2023)**: broader concept affecting **both sexes**, multiple systems (**metabolic, haematological, cardiovascular, psychological, immunological, growth and development, performance**).
- **Low energy availability: under about 30 kcal/kg fat-free mass/day.**
- **Bone: low BMD (Z-score under -1.0 in athletes is concerning; under -2.0 = low BMD for age)** (use **Z-scores**, not T-scores, in premenopausal women and young athletes), **stress fractures (trabecular sites: femoral neck, sacrum, pelvis)**.
- **Treatment: multidisciplinary** (sports physician, dietitian, psychologist); **increase energy intake/reduce training**: restoration of menses; **oral contraceptive pills do not restore BMD** (mask amenorrhoea); **transdermal oestrogen** may be used if needed; calcium and vitamin D.

5. Concussion (sport-related)

- **Traumatic brain injury induced by biomechanical forces**: **functional rather than structural** (normal standard neuroimaging); symptoms may evolve over hours.
- **Signs:** loss of consciousness (**under 10%**), **amnesia**, confusion, headache, dizziness, balance problems, visual disturbance, **"just not right"**.
- **Assessment: SCAT6 (Sport Concussion Assessment Tool; Amsterdam 2022 consensus)** for athletes 13 years and older; **Child SCAT6** for 8-12 years; **Concussion Recognition Tool (CRT6)** for non-medical people.
- **"If in doubt, sit them out": no return to play on the same day.**
- **Red flags (immediate removal and emergency referral):** neck pain/tenderness, **double vision**, weakness/tingling in limbs, **severe or increasing headache**, **seizure**, **loss of consciousness**, **deteriorating conscious state**, **vomiting**, increasingly restless/agitated/combative.
- **Return to sport (graduated, stepwise): relative rest for 24-48 hours** (light activity encouraged), then **symptom-limited activity, aerobic exercise, sport-specific exercise, non-contact training drills, full-contact practice (after medical clearance), return to sport**; **at least 24 hours per step**; **children/adolescents: more conservative; return to learn before return to sport**.

- **Second impact syndrome: diffuse cerebral swelling** after a second concussion before recovery from the first: **adolescents**; often fatal.
- **Chronic traumatic encephalopathy (CTE):** tauopathy associated with repetitive head impacts (contact sports).

6. Pre-participation evaluation and cardiac issues

- **Sudden cardiac death in young athletes: hypertrophic cardiomyopathy (commonest in US athletes)**, **arrhythmogenic right ventricular cardiomyopathy (Italy)**, **coronary artery anomalies**, **myocarditis**, **Marfan syndrome (aortic dissection)**, **commotio cordis** (blunt chest impact during repolarisation: **ventricular fibrillation**; **immediate CPR + AED**), **long QT**, **Brugada**, **WPW**.
- **Screening:** history and examination (AHA 14-point) ± **ECG (European/IOC practice)**.
- **Athlete's heart (physiological)**: sinus bradycardia, **first-degree AV block**, early repolarisation, isolated voltage criteria for LVH: **normal ECG variants in athletes (international criteria)**.
- **Exercise-induced bronchoconstriction**, **exertional heat stroke** (**core temperature over 40 C + CNS dysfunction**: **cold-water immersion ("cool first, transport second")**), **hyponatraemia (exercise-associated: over-drinking hypotonic fluids in endurance events)**.
- **Marfan:** avoid contact and isometric sports (aortic root dilatation).
- **Atlantoaxial instability in Down syndrome** (screening for contact sports, Special Olympics).
- **Return to play after infectious mononucleosis: splenic rupture risk (days 4-21)**: **avoid contact sports for at least 3-4 weeks** (until splenomegaly resolves).

7. Anti-doping

Organisation

- **World Anti-Doping Agency (WADA)**: founded **1999** (after the 1998 Tour de France Festina scandal); publishes the **World Anti-Doping Code** and the **Prohibited List** (updated **annually, effective 1 January**).

- **National anti-doping organisations: India: NADA (National Anti-Doping Agency, established 2005)**; **National Dope Testing Laboratory (NDTL), New Delhi**; **National Anti-Doping Act, 2022**.
- **Court of Arbitration for Sport (CAS)**: appeals.

Criteria for inclusion on the Prohibited List (any two of three)

1. **Potential to enhance sport performance.**
2. **Actual or potential health risk to the athlete.**
3. **Violates the spirit of sport.**
4. (Or masks the use of other prohibited substances.)

Prohibited List categories

- **Prohibited at all times (in and out of competition):**
  - **S0 Non-approved substances**,
  - **S1 Anabolic agents** (**anabolic androgenic steroids**: testosterone, **nandrolone**, **stanozolol**; **other anabolic agents: clenbuterol, SARMs, tibolone, zeranol**),
  - **S2 Peptide hormones, growth factors and related substances** (**EPO and erythropoiesis-stimulating agents**, **HIF activators** (e.g. **roxadustat**, **cobalt**), **hCG and LH (in males)**, **growth hormone and its releasing factors**, **IGF-1**, corticotrophins),
  - **S3 Beta-2 agonists** (all prohibited, **except inhaled salbutamol (max 1,600 µg/24 h, not exceeding 600 µg over 8 hours)**, **inhaled formoterol (max 54 µg/24 h)**, **inhaled salmeterol (max 200 µg/24 h)**, **inhaled vilanterol (max 25 µg/24 h)** within therapeutic doses),
  - **S4 Hormone and metabolic modulators** (**aromatase inhibitors** (anastrozole, letrozole), **SERMs** (tamoxifen, clomiphene), **myostatin inhibitors**, **insulin and insulin mimetics**, **meldonium**, **trimetazidine**, **AICAR, GW1516 (PPAR-delta agonists)**),
  - **S5 Diuretics and masking agents** (**furosemide, thiazides, spironolactone, acetazolamide**, **desmopressin**, **probenecid**, **plasma expanders: albumin, dextran, HES**, **mannitol IV**) (**drospirenone, pamabrom, topical dorzolamide/brinzolamide are permitted**),
  - **M1 Manipulation of blood and blood components** (**blood doping: autologous, allogenic, homologous transfusion**; artificial oxygen

carriers),
    - **M2 Chemical and physical manipulation** (**tampering, urine substitution**, **IV infusions/injections over 100 mL per 12-hour period** except during hospital treatment, surgery or clinical investigations),
    - **M3 Gene and cell doping**.
- **Prohibited in competition only:**
    - **S6 Stimulants** (amphetamines, cocaine, **ephedrine (over 10 µg/mL urine)**, **pseudoephedrine (over 150 µg/mL)**, methylhexaneamine, modafinil, **methylphenidate**, strychnine) (**caffeine, nicotine, phenylephrine, synephrine are on the monitoring programme, not prohibited**),
    - **S7 Narcotics** (morphine, oxycodone, fentanyl, methadone, pethidine, **tramadol (prohibited in competition since 2024)**) (**codeine permitted**),
    - **S8 Cannabinoids** (**except cannabidiol (CBD)**; THC urinary threshold **150 ng/mL**),
    - **S9 Glucocorticoids** (**all injectable routes, oral, rectal** prohibited in competition; **inhaled, topical, intranasal are permitted**; **washout periods** for injections, e.g. about 3 days intramuscular/intra-articular for most).
- **Prohibited in particular sports: P1 Beta-blockers** (**archery, shooting (also out of competition)**, golf, billiards, darts, motorsport, skiing (some disciplines), underwater sports).
- **Local anaesthetics (lidocaine), NSAIDs, paracetamol, insulin for diabetes (with TUE), vitamins** are not prohibited (insulin requires TUE).

Anti-doping rule violations (ADRVs: 11 under the Code)

- **Presence of a prohibited substance** in a sample (**strict liability: the athlete is responsible for any substance found, regardless of intent**), use/attempted use, **evading/refusing sample collection**, **whereabouts failures (3 within 12 months)**, tampering, possession, trafficking, administration, complicity, prohibited association, **intimidating/retaliating against whistle-blowers**.
- **Sanctions: standard 4 years** for intentional violations involving non-specified substances; **2 years** if not intentional; **reduced for "no significant**

fault" or **contaminated products**; **lifetime ban** for repeat serious violations.

- **Athlete Biological Passport (ABP): longitudinal monitoring of biomarkers** (haematological module: haemoglobin, reticulocytes, **OFF-score**; steroidal module: **testosterone/epitestosterone (T/E) ratio (over 4 suspicious)**; endocrine module (GH)); detects doping indirectly.
- **Testing methods:** urine and blood; **isotope ratio mass spectrometry (IRMS: exogenous testosterone)**, **GC-MS/LC-MS**, **dried blood spots**.

Therapeutic Use Exemption (TUE)

- **Criteria (all must be met):**
  - **Significant health impairment** without the substance,
  - **No more than return to normal health** (no additional performance enhancement),
  - **No reasonable permitted therapeutic alternative**,
  - Not a consequence of prior non-therapeutic use of a prohibited substance.
- **Apply in advance** (retroactive TUE for emergencies). Examples: **insulin (diabetes)**, **glucocorticoids (systemic)**, **methylphenidate (ADHD)**, **testosterone (genuine hypogonadism)**, **beta-blockers in shooting (cardiac indication)**.

Doctor's responsibilities

- **Check every medication against the current Prohibited List** (e.g. **Global DRO**, NADA app), **record medications**, **advise on supplements (contamination risk: anabolic agents/stimulants in supplements)**, **apply for TUE**, **IV infusion limits**; **doctors can be sanctioned** (administration, complicity).

8. Other sports medicine topics

- **Platelet-rich plasma (PRP)**: autologous platelet concentrate (**growth factors: PDGF, TGF-beta, VEGF**); **not prohibited by WADA** (growth factors within PRP: **permitted since 2011**); evidence best for **knee OA and lateral epicondylitis** (mixed).

- **Exercise prescription: WHO 2020: adults 150-300 minutes moderate or 75-150 minutes vigorous aerobic activity per week + muscle-strengthening 2 or more days/week**.
- **Athletes and NSAIDs:** may impair **bone and tendon-bone healing** (fracture non-union concern), **renal injury with dehydration**.
- **Sports drinks and carbohydrate loading**; **creatine** (not prohibited; increases phosphocreatine stores: short-burst power).

Exam pearl: **WADA (1999) Prohibited List annual (1 January); India: NADA (2005), NDTL, National Anti-Doping Act 2022. Inclusion = any 2 of 3: performance, health risk, spirit of sport. Strict liability. Standard intentional sanction 4 years. Whereabouts: 3 failures in 12 months.**

Exam pearl: **At all times: anabolic agents, EPO/HIF activators, GH, beta-2 agonists (except inhaled salbutamol under 1,600 µg/day, formoterol, salmeterol), aromatase inhibitors/SERMs, insulin, meldonium, diuretics/masking agents, blood doping, IV infusion over 100 mL/12 h, gene doping. In competition: stimulants, narcotics (tramadol since 2024; codeine allowed), cannabinoids (CBD allowed), systemic/injected glucocorticoids. Beta-blockers in archery/shooting. TUE: health impairment, no enhancement, no alternative.**

Exam pearl: **High-risk stress fractures: femoral neck tension side, anterior tibia (dreaded black line), navicular, Jones (5th MT proximal), medial malleolus, sesamoids. RED-S: use Z-scores; OCP does not restore BMD. Concussion: no same-day return, SCAT6, stepwise return at least 24 h per step. Commotio cordis: AED.**